Cybersecurity for Next-Generation Road Transportation: A Review

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Emerging transportation technologies, including electric vehicles (EVs), autonomous vehicles (AVs), and connected vehicles (CVs), are poised to revolutionize mobility and logistics through advancements in connectivity, automation, and electrification. However, the convergence of these systems introduces substantial cybersecurity challenges, including data breaches, spoofing, and infrastructure attacks, due to increasingly complex and interconnected attack surfaces. This article presents a comprehensive survey of cybersecurity threats, vulnerabilities, and evaluation practices within the AV, CV, and EV landscape. We analyze 227 peer-reviewed studies published between 2021 and 2025 and introduce two novel taxonomies: a three-stage cyber-attack lifecycle framework—Stream to Information (S2I), Information to Decision (I2D), and Decision to Actuation (D2A)—and 11 representative attack paths. Our findings reveal that the stream and data processing stages are most frequently exploited, with limited standardization in evaluation metrics and inadequate emphasis on real-world operational consequences. We highlight the growing importance of integrating quantum-safe cryptography and AI-driven anomaly detection as proactive defense mechanisms. Finally, we offer stage-wise design recommendations and identify future research directions, including the need for cross-domain evaluation frameworks, cyber-social risk assessments, and secure integration of emerging multi-modal systems. This survey aims to support policymakers, researchers, and industry stakeholders in developing resilient, secure, and trustworthy next-generation road transportation ecosystems.

CCS Concepts: • **General and reference → Surveys and overviews;**

Additional Key Words and Phrases: Electric vehicle, connected vehicle, autonomous vehicle, cyber-physical system, cybersecurity, review

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1 Introduction

The next generation of road transportation is defined by electrification, automation, and better connectivity [65]. **Electric vehicles (EVs)** made up about 18% of global new car sales in 2023, a 35% increase from 2022 [2]. By 2040, EVs are projected to represent 10–70% of the global vehicle fleet and 13–100% in the U.S. by 2050 [83, 115]. The transition to EVs presents significant charging infrastructure challenges, while **autonomous vehicles (AVs)** and **connected vehicles (CVs)** are simultaneously advancing as distinct yet complementary smart mobility technologies that promise to reshape future road transportation systems [5]. AVs have undergone extensive pilot testing on public roads for years,[1] demonstrating consistent progress in autonomy and safety [59]. Commercial AV applications have also emerged, such as Tesla's Autopilot and **Full Self-Driving (FSD)** systems[2] corresponding to higher driving automation levels as defined by SAE J3016 [27]. Meanwhile, CVs leverage real-time communication with surrounding infrastructure and vehicles, emphasizing low-latency data exchange [150]. Several pilot programs have demonstrated CVs' potential to enhance traffic efficiency and safety [64].

[1] https://www.nhtsa.gov/automated-vehicle-test-tracking-tool
[2] https://www.tesla.com/support/full-self-driving-subscriptions

Collectively, AV, CV, and EV technologies are converging to revolutionize how we travel and move goods, promising safer and more efficient roads through their combined innovations in electrified powertrains, automation, and connectivity [158]. The need for an integrated perspective has become urgent because the underlying technologies are rapidly converging. First, most early commercial AVs are battery-electric, because drive-by-wire control and regenerative braking are easier to implement on electric drivetrains, so an attack on battery-management firmware can directly compromise autonomous-driving safety [37]. Second, AVs and EVs alike depend on high-bandwidth **vehicle-to-everything** (**V2X**) links for **over-the-air** (**OTA**) updates and smart-charging coordination; a single vulnerability in the ISO 15118 can propagate across connectivity and autonomy layers, as recent large-scale scans of charging back-ends demonstrate [14]. Third, manufacturers are moving toward software-defined vehicle architectures that co-locate propulsion, perception, and communication on common high-performance controllers, enabling cross-layer exploits such as **man-in-the-middle** (**MitM**) firmware injection via the charging interface that disables autonomous-braking functions [154]. This convergence justifies treating AVs, CVs, and EVs together as the "next-generation" road transport paradigm. In contrast, other emerging modes of transport (such as urban air mobility or e-bikes) operate in different domains and scales, and they lack the integrated vehicle–infrastructure ecosystem of on-road automobiles; hence, our focus remains on AVs, CVs, and EVs as the core of next-generation road transportation. Accordingly, we restrict our analysis to AVs, CVs, and EVs, omitting modes without comparable vehicle–infrastructure interaction.

Despite their promises, AVs, CVs, and EVs also introduce significant cybersecurity vulnerabilities that can be exploited by malicious attackers, making cybersecurity a crucial aspect of overall system performance [139]. These vehicles are essentially complex cyber-physical systems with an expanded digital attack surface, stemming from their increased reliance on software, sensors, communication networks, and **machine learning** (**ML**) models. Each technology brings unique attack vectors. For example, EVs can be exposed to attacks on their charging infrastructure and **vehicle-to-grid** (**V2G**) communications, which could enable energy theft [28] or even destabilize portions of the power grid [46]. AVs, on the other hand, rely on an array of sensors (e.g., cameras, LiDAR, Radar), sophisticated sensor-fusion and decision-making algorithms. They may be susceptible to spoofing or adversarial inputs that disrupt their perception [164], control capabilities [148], and navigation [170] (e.g., perturbing road sign images or **global positioning system** (**GPS**) signals to mislead the vehicle's computer vision and path planning modules). Similarly, CVs that depend on inter-vehicle communication face threats like MitM or message falsification attacks on V2X messages. An attacker who injects false traffic data or alters safety messages can create chaos; distorted traffic information could trigger road crashes [97] or gridlock by misleading human drivers and traffic operators [25].

Each of the AV, CV, and EV domains has distinct weaknesses, from battery management systems and electric powertrain signals in EVs to sensor spoofing and **artificial intelligence** (**AI**) manipulations in AVs, and wireless communication protocols in CVs, there is considerable overlap in the underlying electronic architecture of these vehicles. All three rely on networked **electronic control unit** (**ECU**), vehicular communication buses, and external connectivity, meaning certain exploits (such as malware in the infotainment or telematics unit) could affect any modern vehicle regardless of powertrain or autonomy [41, 161]. This framework highlights the importance of adopting a unified cybersecurity strategy, facilitating comprehensive defense mechanisms to address common threats rather than addressing AVs, CVs, and EVs individually.

Real-world incidents highlight the critical need to secure next-generation vehicles. These are not just theoretical vulnerabilities since successful exploits have been demonstrated by researchers. A famous example is the 2015 Jeep Cherokee hack [52], in which Miller and Valasek [108] remotely

infiltrated a Jeep's Uconnect infotainment system over the cellular network and ultimately sent commands to the **controller area network** (**CAN**) bus to disable the engine and brakes while the vehicle was in motion. This dramatic demonstration of a wireless takeover (which caused the Jeep to be stopped on the road) prompted a major recall of 1.4 million vehicles to patch the software [51]. In another example, researchers at Tencent's Keen Security Lab discovered a chain of vulnerabilities in the Tesla Model S (2016) that allowed them to remotely control the vehicle's steering and braking [121]. By exploiting the Tesla's web browser and firmware update channels, they gained unauthorized access to the CAN bus, demonstrating an attack that could engage the brakes or turn the steering wheel from a distance. These incidents illustrate that cyberattacks on modern vehicles can directly translate into loss of control, collisions, or road safety hazards, not to mention financial and privacy harms. They serve as cautionary tales and motivate robust cybersecurity measures for AVs, CVs, and EVs before such attacks migrate from controlled lab environments into the wild.

Recognizing these cybersecurity challenges, industry and government organizations have begun developing frameworks and standards to ensure the safe introduction of AVs, CVs, and EVs. For instance, the new ISO/SAE 21434:2021 standard for road vehicle cybersecurity engineering provides a comprehensive framework for **threat analysis and risk assessment** (**TARA**) throughout the vehicle lifecycle [161]. ISO/SAE 21434 formalizes processes for manufacturers and suppliers to identify vulnerabilities, implement risk-mitigation strategies, and continuously monitor security post-deployment. In the U.S., the **National Highway Traffic Safety Administration** (**NHTSA**) has issued "Cybersecurity Best Practices for Modern Vehicles (most recently updated in 2022)" that endorse ISO/SAE 21434 as an industry best-practice benchmark [120]. Our survey aligns with these frameworks by identifying specific vulnerabilities and attack scenarios that standards like ISO/SAE 21434 aim to guard against, and by highlighting gaps or emerging issues that may inform future revisions of such guidelines. Thus, we bridge the gap between high-level industry standards and the detailed academic literature; the standards set out what should be done for automotive cybersecurity, while our study surveys how the research community is addressing the problem and where further work is needed.

Although numerous studies address individual AVs, CVs, and EVs domains, the literature still lacks an integrated synthesis of cross-domain threats and defenses. Most existing surveys focus on one domain at a time. For example, there are surveys dedicated to AV security, reviewing attacks on sensors, perception, and control of self-driving cars [36, 89, 143]. Separately, other works target CV networks, examining V2X communication protocols and **vehicular ad-hoc network** (**VANET**) attacks [47, 96, 117, 136]. Likewise, the cybersecurity of EVs and charging infrastructure has been surveyed in isolation [6, 43, 57, 114].

Prior surveys are constrained by (i) lack of a unified taxonomy, (ii) limited focus on real-world impacts, and (iii) insufficient analysis of evaluation methodologies. First, we differentiate ourselves by offering a holistic taxonomy that spans all three facets of next-generation road transport. We consider AV, CV, and EV systems together and map out their combined attack surface, whereas prior surveys typically omit cross-domain attack vectors (for instance, an attack that exploits connectivity to impact an autonomous driving function). By examining these technologies collectively, we capture both their distinct vulnerabilities and their common underlying infrastructure, providing a more comprehensive picture of next-generation road transportation's cybersecurity than any single-domain review. Table 1 summarizes related surveys and reveals that none cover AV, CV, and EV cybersecurity in an integrated manner, a gap that our survey explicitly fills.

Second, existing surveys often give limited attention to the real-world impacts of cyberattacks, focusing mostly on technical attack mechanisms. Many survey papers enumerate how an attack can be executed (e.g., by altering sensor data or injecting CAN messages) without fully exploring

Table 1. Related Survey Articles

	AV attack classification	CV attack classification	EV attack classification	Real-world effects	Evaluation platform
[47]		✓			
[57]			✓		
[89]	✓				
[151]	✓	✓			
[96]		✓			
[43]			✓		
[98]	✓	✓			
[136]		✓			
[143]	✓				
[1]	✓	✓			✓
[20]		✓	✓		
[48]	✓				
[114]			✓		
[113]	✓	✓			
[60]	✓	✓			
[144]		✓			✓
[36]	✓				
[6]			✓		✓
[117]		✓			✓
This survey	✓	✓	✓	✓	✓

the consequences if such an attack succeeds in a real deployment. As a result, an attack might be described abstractly as "altering data" [3] or "deceiving a ML model" [128], with little discussion of what that means for vehicle behavior, passenger safety, or traffic operations. This is a critical gap from a safety engineering perspective. It matters greatly whether an attack merely triggers a dashboard warning versus causing a multi-vehicle accident. Our survey specifically addresses this gap by analyzing the real-world effects of attacks. For each category of vulnerability, we review and synthesize findings on the potential physical and operational outcomes, for instance, could the attack lead to loss of braking, route deviations, traffic jams, or hazards to pedestrians? By emphasizing these impacts, we help assess the true risks associated with each attack vector and thereby provide clearer priorities for defense. This focus complements industry safety standards by linking technical vulnerabilities to safety outcomes, which is essential for risk assessment frameworks like ISO 21434's TARA process [161]. Ultimately, highlighting real-world consequences ensures that cybersecurity is not treated as an abstract information technology problem but as a core component of functional safety in road transportation systems.

Third, there is a lack of surveys that examine the methodologies for evaluating the cyberattack impacts. As the body of literature continues to grow, it becomes increasingly important to understand and reflect on the common and appropriate approaches to cybersecurity research. Given the interdisciplinary nature of this field, different research communities may adopt varying standards and criteria. By systematically synthesizing information on current evaluation methods, platforms, and metrics, we can gather profound knowledge about the landscape and work toward establishing more consistent and rigorous evaluation practices.

In this article, we present a new survey to fill the aforementioned gaps by addressing the following research questions:

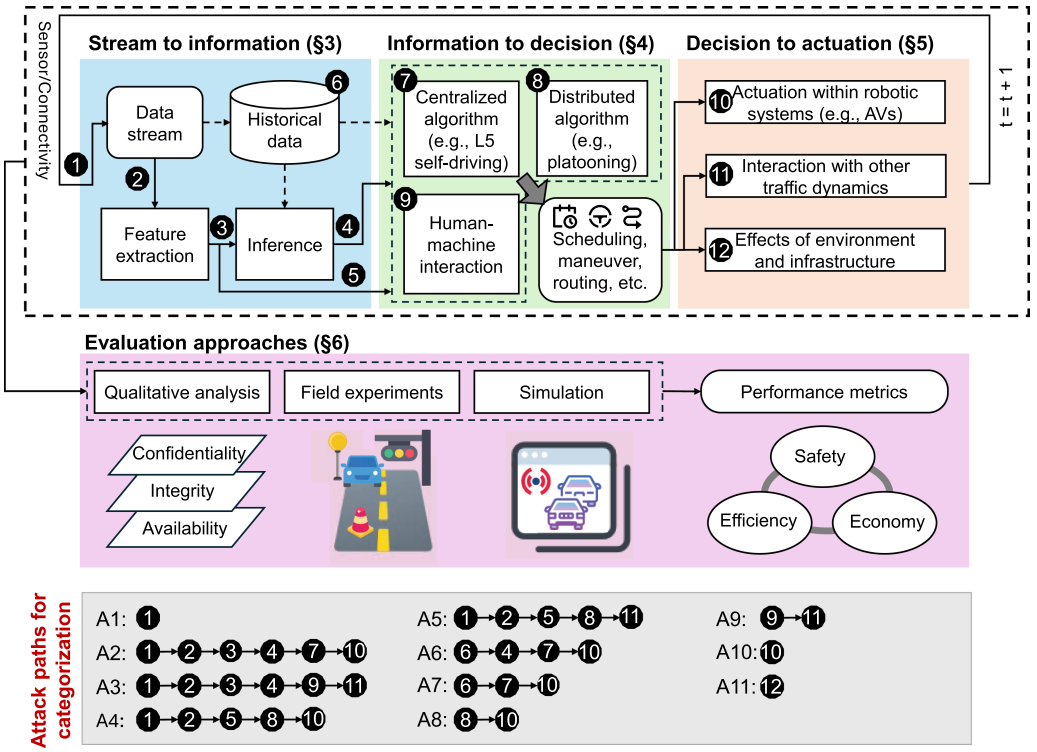


Fig. 1. Survey framework. We group the attacks based on the three-stage framework of road transportation applications for AV, CV, and EV, as illustrated in the top panel, to provide a clear understanding of how threats evolve throughout the system lifecycle. Then, we follow the mid-panel to review evaluation approaches along with a summary of the corresponding performance metrics. Alongside this process, we synthesize attack paths, where each path represents a set of attacks exploiting the same vulnerable components. These paths provide practical value by helping researchers identify recurring threat patterns and enabling manufacturers to pinpoint relevant security concerns. Further details are provided in the next section.

Q1: What components in the operational lifecycle of emerging road transportation applications (e.g., AVs, CVs, and EVs) are vulnerable to security risks?
Q2: What real-world impacts can result from attack vectors and adversarial strategies in practical deployment scenarios?
Q3: What methodologies, frameworks, and metrics are currently used to analyze and evaluate cybersecurity in AVs, CVs, and EVs?

To answer these questions, we propose a novel survey framework structured around two newly introduced taxonomies, as illustrated in Figure 1. First, we establish a three-stage feedback loop taxonomy—**Stream to Information** (**S2I**), **Information to Decision** (**I2D**), and **Decision to Actuation** (**D2A**)—to systematically capture the core processes within AV, CV, and EV applications. Second, we identify 11 distinct attack paths derived from existing literature, highlighting how attacker can exploit vulnerabilities within different stages (i.e., stream, information, decision, actuation, and evaluation) and their real-world implications. Through our categorization, we facilitate recognizing commonalities and unique characteristics across different threats, thereby aiding developers and researchers in prioritizing defensive strategies and allocating necessary resources effectively. For each type of attack, we also examine how its performance is evaluated in the literature, identify

gaps in current assessment practices, and provide a set of consolidated guidelines for evaluating both attacks and defenses. Third, based on this structured synthesis, we offer key design recommendations such as adopting sensor redundancy and robust validation mechanisms (S2I), context-aware decision verification (I2D), and resilient actuation and anomaly detection practices (D2A), all of which are crucial for comprehensive security assurance across transportation systems. In addition to these operational security threats, the instrumentation of AVs, CVs, and EVs also raises new privacy concerns. Location tracking and the risks of new perception capabilities (e.g., cameras that are always on) could expose sensitive data about consumers, drivers, and pedestrians. There are several proposals to mitigate these risks, such as data minimization, anonymization, differential privacy, and regulatory compliance. Finally, we are focused on how attacks can affect the operation of vehicles, and we refer the reader to other surveys focusing on privacy [138, 152, 157, 171]

The rest of this article is organized as follows. Section 2 describes our methodology for literature collection, filtering, and synthesis. Sections 3 through 5 present our analysis of cyber-attacks, categorized according to the stages in Figure 1. Within each section, we further organize the discussion based on the emerging road transportation applications i.e., AVs, CVs, and EVs, and the attack paths being utilized in various cyberattacks. Section 6 comprehensively overviews the existing evaluation platforms and metrics used to assess cybersecurity performance in next-generation road transportation. Section 7 discusses the key findings in the different stages of next-generation road transportation applications, implications for the government and industry sector, and gaps in cybersecurity research. We conclude our research and explore promising avenues for future research in Section 8.

2 Data and Methodology

A **Systematic Literature Review** (**SLR**) [90, 129] is conducted in this study to investigate various types of potential cyberattacks, categorization of these attacks within different stages of the workflow of emerging road transportation applications i.e., AVs, CVs, and EVs, and ways of exploiting the vulnerabilities of systems accommodating these applications. We have structured our methodology into four pivotal stages in accordance with established SLR guidelines: (1) formulating research questions, (2) strategizing the literature search, (3) filtering and evaluating the searched literature, and (4) synthesizing the filtered literature and categorizing cyberattacks following a proposed workflow of emerging road transportation applications (see Figure 1) ensuring robust coverage and thoughtful analysis of the current state of research in cybersecurity for next-generation road transportation applications. The set of formulated research questions is presented in the previous section.

2.1 Strategy for Literature Search

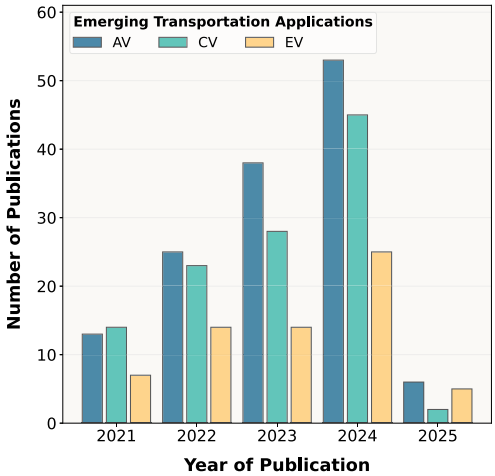
Following the research questions formulated, keywords are selected to perform the literature search in the Scopus database for AVs, CVs, and EVs separately. Table 2 shows an overview of the keywords used for the literature search and the number of scientific documents retrieved for publications between 2021 and 2025. This five-year period was selected due to a significant increase in cybersecurity-related publications focusing on AVs, CVs, and EVs, beginning in 2021. The literature search was performed on 28 March 2025.

2.2 Filtering and Evaluation of Literature

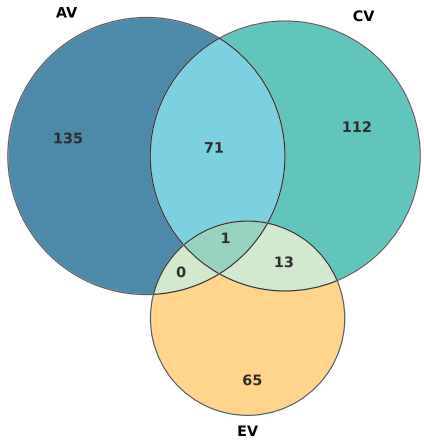
Each study retrieved from the initial search went through a stringent filtering process, initially based on title, abstract, and conclusion, followed by a thorough review of the full text to ensure compliance with the objectives and scope of this research. Following this scrutiny, 207, 197, and 79 documents are filtered out, corresponding to AV, CV, and EV-related studies, respectively.

Table 2. Overview of Literature Search Keywords and Document Counts

Transportation Applications	Keywords used for Literature Search	No. of Scientific Documents Retrieved
AV	("autonomous vehicle") AND ("cyberattack" OR "cybersecurity" OR "cyber threat" OR "cyber risk" OR "cyber vulnerabilities")	1212
CV	("connected vehicle") AND ("cyberattack" OR "cybersecurity" OR "cyber threat" OR "cyber risk" OR "cyber vulnerabilities")	1045
EV	("electric vehicle") AND ("cyberattack" OR "cybersecurity" OR "cyber threat" OR "cyber risk" OR "cyber vulnerabilities")	481



(a) Year-wise distribution of relevant publications from 2021 to 2025.



(b) Overlap among research papers addressing AV, CV, and EV technologies.

Fig. 2. Summary of literature search results based on the Scopus database highlighting temporal trends and technological overlaps.

To illustrate the temporal trends in the filtered literature, a histogram was used to visualize the number of publications related to AVs, CVs, and EVs from 2021 to 2025. This distribution, based on data retrieved from the Scopus database, is presented in Figure 2(a), highlighting the growth in research interest across all three technology domains. In addition to the temporal distribution, a Venn diagram (Figure 2(b)) summarizes the degree of overlap among the retrieved documents, revealing how many publications address more than one technology. Together, these visualizations provide insights into both the evolution and the interdisciplinary nature of research on emerging road transportation applications.

2.3 Synthesizing Attack Paths

We synthesize the attack paths using the three-stage workflow (S2I, I2D, and D2A) introduced in Figure 1. Attack paths convey detailed information about the specific components involved in cyberattacks, going beyond generic threat names like “GPS spoofing” or “MitM”, offering a structured and component-aware view of how attacks manifest in practice. This taxonomy reduces the need for brute-force mapping from abstract threat names to concrete system components when analyzing or defending against attacks. For example, an attack labeled as “GPS spoofing” may affect positioning modules, navigation systems, and downstream decision logic whereas our attack paths make these affected components explicit. This granularity is particularly valuable

for industry stakeholders, such as component manufacturers and system integrators, who can use this information to identify potential exposure and harden specific modules. Additionally, the proposed attack paths can serve as a taxonomy for future attacks, facilitating systematic threat classification. Importantly, by illustrating how individual threats propagate across S2I, I2D, and D2A stages, the attack paths help trace cascading effects and support post-attack diagnostics through component-level analysis.

The potential attack paths with an example cyberattack and their implications are described below:

- **A1 (Remote Sensor Attack)**: An attacker manipulates the data stream entering the workflow.
- **A2 (GPS Spoofing Attack)**: Altered data leads to incorrect feature extraction and flawed predictions, causing centralized algorithms (e.g., Level 5 autonomous driving) to make incorrect decisions, resulting in faulty actuation within robotic systems.
- **A3 (Replaying Attack)**: Similar to A2, but in later stages, human–machine interactions are disrupted due to inaccuracies in predicted data, leading to hazardous interactions with traffic dynamics.
- **A4 (Relaying Attack)**: Manipulated feature extraction misguides distributed algorithms (e.g., vehicle platooning), leading to incorrect decisions and faulty robotic actuation.
- **A5 (Data Falsification Attack)**: Similar to A4, but affects human–machine interactions, leading to unsafe interactions in traffic.
- **A6 (Data Poisoning Attack)**: Manipulating historical data results in inaccurate predictions, causing centralized algorithms to make incorrect decisions and triggering faulty robotic actuation.
- **A7 (Backdoor Attack)**: Altered historical data directly compromises centralized algorithms, leading to erroneous decisions and robotic system failures.
- **A8 (MitM Attack)**: By intercepting communication channels, an attacker alters distributed algorithms, causing incorrect robotic actuation.
- **A9 (Phishing Attack)**: Manipulated interactions disrupt human–machine collaboration, leading to hazardous traffic conditions.
- **A10 (Physical Manipulation – Brake Tampering)**: Direct physical interference affects vehicle actuation (e.g., removing the brake).
- **A11 (Environmental and Infrastructure-Based Attack)**: External factors, such as creating hazardous road conditions (e.g., making roads extremely slippery), disrupt transportation safety.

By understanding these attack paths, researchers and policymakers can develop robust countermeasures to enhance the security and robustness of emerging road transportation technologies.

3 Attacks During the Stream to Information (S2I) Stage

In this article, we categorize cyberattacks based on their core mechanisms, with each subsection discussing two or three specific attacks and their impacts on AVs, CVs, and EVs. These cyberattacks can occur through various communication channels, including **vehicle-to-sensor** (**V2S**), **vehicle-to-vehicle** (**V2V**), **vehicle-to-infrastructure** (**V2I**), V2G, **vehicle-to-network** (**V2N**), and V2X interactions [7, 118, 145].

In the following subsections, the attack types are grouped based on similarities in their execution to facilitate a clearer understanding of their patterns. For each attack mechanism, we provide a brief description and identify the corresponding attack paths (A1 to A7), as illustrated in Figure 1. We then discuss the potential real-world impacts of these attacks, particularly how

Table 3. Cyberattacks During S2I Stage

Attack Type	Cyber Attack	Definition	Attack Paths	Affected Metrics	Related Works
Sensor and Interception	Remote Sensor Attack	Manipulation of AV sensor data during transmission (V2S), misleads vehicles about surroundings.	A2	Safety, Efficiency	[80]
	Eavesdropping	Interception of V2V/V2I/V2N channels to capture sensitive data (e.g., charging transactions, user info).	A1	Safety	[12, 137]
Routing and Location-Based	Routing	Alters packet routes in CVs (V2V/V2I/V2N) to reroute through hostile servers.	A2	Safety	[92, 130]
	Location Trailing	Tracks vehicle location via V2N/V2S, compromising privacy and safety.	A1	Safety	[4]
	GPS Spoofing	Spoofs GPS to mislead vehicle location, used in AVs/CVs/EVs	A2	Safety, Efficiency	[58, 159, 166]
Close Proximity and Replay-Based	Close Proximity and CAN	Exploits short-range systems (e.g., TPMS, RKE) via V2N/V2S to manipulate controls.	A2	Safety, Efficiency	[53, 116, 134]
	Replaying	Replays valid data (e.g., fob or authentication tokens) to deceive systems.	A2, A3	Efficiency	[74]
	Relaying	Inserts attacker between parties, forwarding messages while eavesdropping.	A4	Efficiency	[28]
Data Falsification and Repudiation	Data Falsification	Alters data for incorrect CV decisions; exploits ML-based classifiers.	A5, A6, A7	Safety, Efficiency	[31]
	Repudiation	Denies or alters logs (e.g., EV charging), spoofing sessions or digital signatures.	A2	Safety, Economy	[66]

they may compromise system performance in terms of safety, efficiency, and economic reliability. Additionally, Table 3 summarizes the cyberattacks associated with the S2I stage.

3.1 Sensor and Interception Attacks

Remote sensor attacks and eavesdropping are unified in this category because they both undermine the earliest stages of the vehicular data lifecycle, targeting sensor measurements or communication channels carrying raw or near-raw data. In cybersecurity terms, these attacks compromise confidentiality and integrity by injecting false signals, intercepting legitimate sensor outputs, or exploiting weak encryption in vehicular networks, ultimately distorting a vehicle's perception of its environment. This shared threat model, characterized by adversarial manipulation or disclosure of sensor data prior to higher-level decision-making, necessitates similar mitigation strategies such as robust encryption, authenticated sensor readings, and secure communication protocols. Koscher et al. demonstrated how sensor and communication vulnerabilities in automobiles could be exploited to remotely influence core vehicle operations [91].

Remote Sensor Attack. AVs face threats during the S2I stage primarily through a remote sensor attack. Remote sensor attack involves manipulating sensor data during transmission through the V2S medium. This attack is included in the S2I stage because it directly targets the sensor data being collected by AVs, which is crucial for gathering information required in their decision-making processes. The remote sensor attack affects the control and sensor components, potentially leading AVs to misinterpret altered data and increasing the risk of collisions or accidents. For example, an attacker altering an AV's lidar readings can make the vehicle perceive a clear path as

obstructed. Jin et al. [80] presented an adaptive control framework for CVs operating under sensor and actuator attacks, which incorporates a decision-making module consisting of a controller and a corrective signal, designed to maintain the uniform boundedness of the platoon system.

Remote sensor attacks can exploit A2 attack path by targeting different stages of the data flow. Through A2, attackers compromise feature extraction and interpretation processes, where manipulated or deceptive sensor data leads to flawed situational awareness. This illustrates how remote sensor attacks can propagate from data collection to higher-level decision-making, thereby threatening both the system's safety and operational efficiency.

Eavesdropping. Eavesdropping is a recurring threat in AVs, CVs, and EVs alike. In the context of CVs, eavesdropping involves intercepting communication channels, capturing data transmitted through V2V, V2I, and V2N mediums, affecting control and communications components, and leading to unauthorized access to sensitive information. For example, intercepting communication between a CV and its base station can reveal critical data. Bajracharya [12] investigated how mobility of vehicle, network association, and encryption time affect secure communication in the **Internet of Vehicles** (**IoV**) when exposed to combined eavesdropping and jamming attacks. The results showed that to maintain the same secrecy rate, these attacks impose additional costs in terms of increased transmit power and bandwidth compared to scenarios without such threats.

In the context of EVs and charging infrastructure, eavesdropping cyberattacks can target various communication channels and systems to intercept and gather sensitive information related to EVs, charging transactions, and user data [137]. The attack mechanisms include wireless network eavesdropping, back-end system eavesdropping, charging station-to-charging station communication, mobile app or web portal eavesdropping, location data interception, and V2G communication. These attacks compromise components such as batteries, controllers, and wireless charging systems, posing significant privacy and security risks by exposing personal and operational data.

Eavesdropping attacks leverage the A1 attack path by exploiting vulnerabilities in the initial data acquisition and transmission phases. Through A1, attackers intercept data streams exchanged via V2V, V2I, V2N, and V2G communication channels, enabling unauthorized access to sensitive information without altering the data itself. This passive form of attack compromises the confidentiality of control, communication, and user-related data, which can be further exploited for targeted intrusions or service disruptions. In EVs, such interception extends to charging transactions, battery management systems, and user location data, posing significant privacy and security risks. By operating through A1, eavesdropping attacks undermine trust in vehicle systems and infrastructure, with potential repercussions for both user safety and system integrity.

3.2 Routing and Location-Based Attacks

Routing manipulation and location-based exploits, such as GPS spoofing, are grouped together since they both target the vehicular network's capacity to deliver accurate route and position information. Rather than attacking raw signals at the sensor interface, these adversaries exploit higher-level processes (e.g., packet routing mechanisms or geolocation subsystems) to misdirect vehicles, disrupt navigation, or manipulate location data. The common thread in this threat model is that adversaries compromise the authenticity and integrity of routing or location information, creating cascading failures in decision-making modules that rely on accurate situational awareness. Checkoway et al. remain a study in exposing how vehicular communication channels are susceptible to such network-level and location-centric attacks, highlighting the need for secure routing protocols and spoofing detection [23] .

Routing. CVs are susceptible to various attacks during the S2I stage, among which routing is a primary threat. This attack manipulates data packet pathways by rerouting CV data packets through malicious servers, targeting V2V, V2I, and V2N mediums and the communications

component. This can lead to data leaks or exposure to harmful actions. For example, rerouting a CV's data packets through a hostile server can siphon off sensitive data. Purohit et al. [130] assessed the black hole attacks' impact on the **Ad hoc On-Demand Distance Vector** (**AODV**) routing protocol in VANETs and proposed a **secure vehicular on-demand distance routing** (**SVODR**) protocol to mitigate their effects. Experimental results showed significant degradation of AODV routing protocol performance under black hole attacks, while the proposed SVODR protocol outperforms the **intrusion detection system** (**IDS**) in VANETs, providing a secure and efficient solution for vehicular communication. A secure AODV routing protocol to identify and mitigate black hole attacks in VANETs is developed by Kumar et al. [92]. The proposed protocol, which incorporates cryptography-based encryption and decryption and modifies RREQ and RREP packet protocols, demonstrated improved network performance under black hole attacks compared to the original AODV routing protocol using various network parameters.

Routing attacks utilize the A2 attack path by compromising the integrity of data during its interpretation and routing within the communication infrastructure. While the initial data streams might remain unaltered at the point of sensing (unlike A1 attacks), adversaries exploit vulnerabilities in the feature extraction, routing, and network-level data handling processes. By injecting false routing information or manipulating network control messages, attackers can divert legitimate data packets through malicious nodes or servers. This rerouting disrupts the normal flow of information, enabling data interception, degradation of communication reliability, and potential insertion of malicious payloads. The consequences extend beyond data leaks, as compromised routing can misguide decision-making systems dependent on timely and accurate information exchanges. By operating through the A2 path, routing attacks threaten both the integrity of vehicle communication networks and the safety-critical functions relying on them.

Location Trailing. Location trailing entails the unauthorized tracking of a vehicle's location, and it most commonly appears in the context of CVs. Attackers typically hack into the CV's location data stream using the V2N and V2S mediums, targeting control, sensor, and communications components. The privacy of vehicle users is compromised, potentially exposing them to threats. For instance, an attacker could continuously track a celebrity's CV, revealing their real-time location to paparazzi or potential harm-doers. Alam et al. [4] proposed a technique combining Centroid Localization and Graham Scan's Hull algorithms to recognize attackers and delineate jammed areas in VANETs. It achieves high accuracy in identifying jamming attack scenarios, contributing to improved VANET security.

Location-trailing attacks leverage the A1 attack path by intercepting location data at the point of collection or during transmission. Unlike attacks that alter data, location trailing focuses on passively extracting sensitive positional information as it flows through V2N and V2S links. By compromising the data stream early, attackers can continuously monitor vehicle movements without detection, leading to serious privacy breaches. The A1 path enables such interception before data reaches higher-level modules, posing significant risks to user privacy and, in targeted cases, personal safety.

GPS Spoofing. GPS spoofing is another attack type that appears in multiple contexts. AVs are susceptible to GPS Spoofing attacks during the S2I stage. This attack misleads AV's GPS systems by sending deceptive signals. Broadcasting counterfeit GPS signals through V2I, V2N, and V2S mediums, affects the control, sensor, and communications components. As a result, AVs might be directed to incorrect or dangerous locations. A notable scenario is an AV misled by spoofed GPS signals into an unsafe area. Vitale et al. [159] dealt with improving the security as well as privacy of communication in CV and AV systems. This study introduced an architecture called "CARAMEL", which assesses the integrity of vehicle-transmitted information and detects GPS spoofing attacks using multi-radio access technology, a Multi-Access Edge Computing

platform, an intelligent onboard unit, and a Public Key Infrastructure. Yang et al. [166] proposed a identification methodology for GPS spoofing attacks on AVs and CVs, utilizing transportation and vehicle engineering knowledge. The proposed method involves a computationally efficient driving model learned from historical vehicle trajectories to predict normal behaviors and detect anomalies. In the case of EVs and EV charging infrastructure, spoofing cyberattacks exploit weaknesses in authentication, communication protocols, and validation mechanisms to deceive the system or users [58]. Some attack mechanisms specific to EVs include charging station spoofing, EV identity spoofing, EV authentication spoofing, charging infrastructure spoofing, GPS spoofing for EV navigation, **Wireless Fidelity** (**Wi-Fi**) and cellular network spoofing, identity spoofing for EV software updates, and EV manufacturer identity spoofing. Such an attack can utilize any medium to target any component of the EV system.

GPS spoofing attacks operate through the A2 attack path by compromising the interpretation of received data during feature extraction and situational awareness processes. Unlike attacks that eavesdrop raw sensory inputs (A1), GPS spoofing deceives the system by feeding false positional data into the perception and decision-making layers. This misguidance propagates through navigation, routing, and control algorithms, leading to faulty actuation decisions. By exploiting the A2 path, GPS spoofing undermines both the operational safety of vehicles and the transportation systems' efficiency reliant on accurate location data.

3.3 Close Proximity and Replay-Based Attacks

Close proximity exploits and replay-based attacks reside under one group because they both harness preexisting, legitimate signals (short-range or internal communication channels) to gain unauthorized access or functionality. This threat model relies on an adversary's ability to either physically approach the vehicle or intercept localized signals (e.g., CAN bus messages, key fob transmissions) and replay or manipulate them in real time. Attackers bypass typical authentication or session expiration safeguards by reusing valid credentials or low-level frames. Miller and Valasek [108] showcased the real-world implications of such short-range and internal network exploits in unmodified production vehicles, underscoring the urgent need for encryption, robust authentication, and IDSs specifically tailored to near-field vehicular protocols.

Close Proximity and CAN Vulnerabilities. Close proximity and CAN vulnerabilities involve exploiting a vehicle's short-range communication systems. Attackers intercept or manipulate communications within close range, predominantly through V2N and V2S mediums, affecting the control, sensor (e.g., **tire pressure monitoring system** (**TPMS**) and **remote keyless entry** (**RKE**)), and communications components (e.g., Bluetooth). The attack provides unrestricted access to primary vehicle functions. Rouf et al. [134] revealed privacy concerns and the lack of authentication in TPMS, allowing for remote spoofing of sensor messages. Ref. [116] identified vulnerabilities in the SAE J1939 specification that can be exploited for impersonation and DoS attacks, and proposed an authentication mechanism as a countermeasure, demonstrating that its introduction does not significantly affect bus communication reliability. For enhancing the existing rolling-code-based RKE systems' security against replay attacks, Greene et al. [53] demonstrated the vulnerability of RKE systems to replay attacks using LimeSDR, which manipulates wireless signals. In order to improve the RKE systems' security, a timestamp-based defense mechanism integrated with the rolling code is proposed and implemented by them.

These attacks exploit the A2 attack path by compromising data interpretation and command execution processes within the vehicle's internal network. Attackers in close proximity exploit weaknesses during feature extraction and command dissemination, injecting falsified or replayed data into the system. This misguides decision-making logic and actuator commands, leading to

unauthorized control over vehicle functions. By operating through A2, close proximity and CAN attacks threaten both the safety and functional integrity of vehicle operations.

Replaying. This involves capturing and retransmitting data packets to deceive CV systems. Signals are intercepted, stored, and replayed using V2V, V2I, and V2N mediums, primarily targeting communications. The attack can lead to unauthorized access or deceptive actions on the CV. An example is the interception and replay of signals from a CV's key fob by a thief to unlock the vehicle. Iqbal et al. [74] modeled typical road scenarios using the Eclipse MOSAIC simulation framework and introduced replay and bogus information cyberattacks in VANETs. The model illustrates the effects of these attacks and supports the advancement of ML algorithms for anomaly detection and mitigation in CVs. Replaying cyberattacks occur on EVs and EV charging infrastructure through exploiting vulnerabilities in communication channels and protocols, allowing attackers to reuse intercepted data to deceive the system or gain unauthorized access. The attack mechanisms specific to this type of cyberattack include authentication token replay, command replay, payment transaction replay, charging data manipulation, data integrity attacks, session hijacking, and OTA update replay. This attack can leverage any medium to target all of the EV components. Replay attacks have also been identified in AV scenarios, in the context of key fob signal interception and retransmission.

Replay attacks exploit both the A2 and A3 attack paths, depending on their point of intervention. Through A2, attackers target data interpretation and processing stages, injecting previously captured packets into network communications to mislead decision-making systems. In contrast, A3 represents attacks where replayed data directly interferes with human–machine interactions, disrupting user commands or system responses. For instance, command replay can trick user interfaces or downstream systems into executing unauthorized actions. By leveraging these paths, replay attacks compromise system integrity, reduce operational efficiency, and pose significant security threats.

Relaying. In a relaying cyberattack, the adversary captures and forwards communication between two entities without alerting them, effectively impersonating each side to the other [28]. The attacker positions itself between the legitimate communicating parties and forwards messages or data between them, making it appear as if the communication is direct and secure when, in reality, it is being intercepted and manipulated by the attacker. Relaying attacks are particularly concerning because they allow attackers to access confidential data, and assume the identities of trusted entities in the communication process. Such attacks can leverage V2V and V2I mediums to hamper the controller and wired charging components of EVs. They are similarly relevant for other vehicle types that depend on authentication exchanges.

Relaying attacks operate through the A4 attack path, which targets the feature extraction and distributed decision-making processes. Unlike simpler interception attacks, relaying involves real-time transfer of in-transit data after initial reception, misleading distributed algorithms (e.g., vehicle platooning, cooperative driving systems) by providing altered or delayed information. This results in erroneous local decisions and faulty actuator responses. The attacker's ability to maintain a seemingly valid communication flow while relaying critical information degrades both the efficiency and reliability of vehicle systems. By exploiting A4, relaying attacks present significant challenges to maintaining secure and trustworthy distributed vehicular operations.

3.4 Data Falsification and Repudiation Attacks

Data falsification and repudiation threats share a focus on undermining the reliability and traceability of higher-level data artifacts ranging from ML inputs in CVs to billing records in EV charging transactions. Adversaries in this category manipulate the content, origin, or authenticity of data so that downstream decision-making processes or forensic analyses cannot distinguish fraudulent

information from legitimate records. The underlying cybersecurity principle uniting these attacks is that the system's trust model hinges on verifiable data integrity, which falsification and repudiation both erode by injecting deceptive information or tampering with evidence of authenticity. Woo et al. [162] comprehensively analyzed how message modification and authentication loopholes can critically affect automotive networks, demonstrating the vital need for cryptographic protocols and secure logging to counteract data falsification and repudiation.

Data Falsification. In the S2I stage, CVs are susceptible to attacks that aim to manipulate or disrupt the processing of collected data and the generation of useful information based on that data. These attacks can result in incorrect or hazardous actions by CV systems, thereby compromising the safety and efficiency of the vehicles. The following are the primary attacks that pose risks to the decision-making processes of CVs. This involves tampering with data to mislead CV systems. Attackers inject or modify data in CVs through V2V, V2I, and V2N mediums, targeting control and communications components. CVs may make incorrect decisions based on this falsified data, leading to potential hazards. An instance is an attacker feeding false sensor data to a CV, causing unnecessary swerving, which risks collisions. Cui et al. [31] discussed the vulnerability of deep learning models used in connected AVs to data poisoning attacks. This study develops a black-box, clean-label data poisoning attack method by employing a hybrid optimization technique, integrated with particle swarm optimization with simulated annealing. The proposed method was tested on traffic sign recognition systems in CAVs and was found to significantly degrade the target models' classification accuracy using only a small fraction of poisoned data samples. Further, Mia and Amini [107] explored data poisoning attacks and their impact on computer vision algorithms for AVs. It also developed a defense mechanism to secure object detection against such attacks.

Data falsification attacks exploit the A5, A6, and A7 attack paths depending on the stage and method of intervention. Through A5, attackers tamper with real-time feature extraction, misleading human–machine interactions and immediate system responses. A6 involves poisoning historical data used for model training, resulting in systematic prediction errors and flawed centralized decision-making. A7 further escalates this by embedding backdoors within training data, directly compromising core algorithms and causing persistent system failures. Collectively, these attack paths demonstrate how falsified data can propagate through perception, learning, and decision-making layers, severely impacting both the safety and efficiency of vehicle operations.

Repudiation. Repudiation attacks typically arise in EVs and **EV charging stations** (**EVCSs**) but can affect other connected systems as well. In the EV domain, repudiation can target various components and communication channels within the EV ecosystem [66]. The attack mechanisms may include transaction tampering, charging session spoofing, manipulating charging infrastructure logs, digital signature forgery, identity impersonation, communication interception, back-end system tampering, and unauthorized access to user accounts. Repudiation can attack wired and wireless charging components through V2I and V2N channels. By undermining transaction verifiability, these attacks threaten both consumer trust and the integrity of the charging process.

Repudiation attacks utilize the A2 attack path by targeting data interpretation and verification processes after initial transmission. Unlike interception at the sensing level (A1), repudiation attacks alter or inject falsified data during validation stages, deceiving systems into accepting manipulated records or unauthorized commands as legitimate. This can lead to financial fraud, unauthorized resource access, and compromised user privacy. By exploiting A2, repudiation attacks disrupt the economic integrity of EV ecosystems and pose significant risks to user and system safety.

Table 4. Cyberattacks During I2D Stage

Attack Type	Cyber Attack	Definition	Attack Paths	Affected Metrics	Related Works
Impersonation and MitM	Impersonation	Fakes legitimate V2X services, installs malicious data/software updates.	A8	Safety	[10]
	Man in the Middle (MitM)	Intercepts/modifies data between vehicle systems using wireless vulnerabilities.	A8	Safety, Efficiency	[67]
Sybil and Free-Ride	Sybil	Creates fake/stolen identities in EVCS to hijack priorities or overload.	A8	Efficiency	[42]
	Free-Ride	Uses shared credentials or hijacks charging sessions in EVCS.	A8, A9, A10	Economy	[17, 79]
Malware and Phishing/Social Engineering	Malware Injection	Injects malicious firmware/software into vehicle systems.	A8, A10	Safety, Efficiency	[61, 163]
	Phishing / Social Engineering	Tricks users via fraudulent messages to steal credentials or install malware.	A9	Safety, Economy	[149]

Summary of this section.

— During the S2I stage, emerging road transportation applications, including AVs, CVs, and EVs, are vulnerable to remote sensor attacks, eavesdropping, routing, location trailing, GPS spoofing, close proximity and CAN vulnerabilities, replaying, relaying, data falsification, and repudiation.

— These attacks exploit various communication mediums (e.g., V2V, V2I, V2N, V2S) and target critical components such as sensors, controllers, communication modules, and charging interfaces. The consequences include unauthorized data access, manipulation of sensor and communication data, privacy violations, disruption of decision-making processes, and degradation of safety, efficiency, and economic reliability.

4 Attacks During the Information to Decision (I2D) Stage

This section focuses on cyberattacks that target the decision-making processes of AVs, CVs, and EVs during the I2D stage. We cluster these attacks based on similarities in their underlying mechanisms to illustrate how adversaries can manipulate or disrupt critical decision-making processes across various vehicular systems. A brief description of the attack mechanism is provided for each attack, and the corresponding attack paths (A8 to A10) are identified, as illustrated in Figure 1. We then examine the potential real-world impacts, particularly how such attacks may compromise system performance in terms of safety, efficiency, and economic reliability. Table 4 summarizes the cyber threats associated with the I2D stage.

4.1 Impersonation and Man in the Middle Attacks

Impersonation and MitM attacks are closely related in that both exploit trust relationships within vehicular communication networks. These attacks target authentication and data integrity by inserting malicious actors into trusted communication flows—either by mimicking legitimate entities or intercepting data exchanges. Grouping them together highlights their shared reliance on compromised identity and communication channels, and their potential to undermine system reliability across AV, CV, and EV environments.

Impersonation. Impersonation attacks occur when unauthorized entities mimic legitimate CV components or services through V2V, V2I, and V2N mediums, primarily affecting the communications component. This can result in CVs downloading malicious data or software,

causing malfunctions or unauthorized control, such as an attacker setting up a fake software update server to distribute harmful software. To identify and avoid malicious vehicles, Appathurai et al. [10] focused on data injection and impersonation attacks in **field-programmable gate array** (**FPGA**)-based IoV and propose a **Washable Physical Unclonable Function** (**WPUF**)-based authentication scheme. The results demonstrate that the proposed WPUF-based authentication scheme is more efficient in verifying tokens compared to conventional techniques.

Impersonation attacks exploit the A8 attack path by directly compromising distributed algorithms and network-level authentication processes. Through A8, attackers insert themselves into trusted communication flows, deceiving vehicles into accepting spoofed identities or malicious commands as legitimate. This bypasses basic integrity checks and leads to unauthorized access, manipulation of vehicle functions, and disruption of system safety. By leveraging A8, impersonation attacks pose critical risks to the security and reliability of CV operations.

Man in the Middle. MitM cyberattacks on EVs and charging infrastructure exploit vulnerabilities in communication channels and protocols to intercept and manipulate data exchanged between different components within the EV ecosystem [67]. Some attack mechanisms specific to this domain include Wi-Fi and cellular network interception, insecure communication protocols, impersonation and spoofing, rogue charging stations, software update tampering, supply chain attacks, and exploiting Bluetooth and **near-field communication** (**NFC**) vulnerabilities. MitM attacks can leverage any of the mediums to hamper battery, controller, and wired charging components. Although primarily illustrated here for EV scenarios, MitM techniques are also relevant to AVs and CVs that rely on similar communication protocols.

MitM attacks leverage the A8 attack path by inserting attackers into distributed communication flows, allowing them to intercept, alter, or inject malicious data while maintaining the illusion of secure exchanges. Unlike passive eavesdropping, MitM involves active manipulation of ongoing communications, deceiving systems into acting on falsified data or executing unauthorized commands. By exploiting A8, MitM attacks undermine system safety, compromise transaction integrity, and negatively impact both operational reliability and economic performance across AV, CV, and EV systems.

4.2 Sybil and Free-Ride Attacks

Sybil and Free-Ride attacks are grouped together due to their shared exploitation of identity and access control vulnerabilities within CV and EV systems. Both attack types involve deception through false or unauthorized identities to gain illegitimate access to resources or influence system behavior. By undermining authentication, resource allocation, and economic fairness, these attacks erode system trust and efficiency, particularly within EV charging infrastructure and networked vehicular environments.

Sybil. During the I2D stage, Sybil attacks involve generating numerous fake identities to disrupt remote actions, creating confusion within the system. These attacks can compromise financial stability by enabling unauthorized access and leaking credentials, ultimately undermining user trust and data privacy. In the context of EVs and charging infrastructure, a Sybil cyberattack targets vulnerabilities in communication and control systems to manipulate network decisions, disrupt charging operations, or obtain unauthorized access to resources [42]. Some of the attack mechanisms of a Sybil cyberattack on EVs and charging infrastructure include network manipulation, manipulating charging sessions, resource exhaustion by overwhelming charging infrastructure, impacting charging priorities or energy distribution, fraudulent charging activities, overcharging and disruption, data manipulation, and privacy violation. Sybil attacks target wireless charging components by utilizing V2V, V2I, and V2N mediums.

Sybil attacks utilize the A8 attack path by deceiving distributed algorithms through the injection of fabricated identities. By overwhelming the system with bogus nodes or entities, attackers distort network-level decisions, misguide resource allocation, and disrupt cooperative vehicular operations. This manipulation degrades operational efficiency, affects system stability, and erodes user trust. Through A8, significant threats to the integrity and reliability of EV ecosystems and other CV systems are posed by Sybil attacks.

Free-Ride. In the context of EV systems, free-ride cyberattacks can target both the vehicles themselves and the charging infrastructure [17, 79]. The attack mechanisms of this type of cyberattack may include Wi-Fi network piggybacking, session hijacking, physical access to charging stations, credential sharing, unintentional access sharing, and password reuses. Free-ride can attack wireless charging by leveraging the V2I medium.

Free-ride attacks can exploit the A8, A9, and A10 attack paths, each representing a distinct attack mechanism. Through A8, attackers impersonate legitimate users or vehicles to gain unauthorized access. A9 involves phishing-based deception, where users are tricked into revealing credentials or approving illegitimate charging sessions. A10 represents physical manipulation, where attackers exploit direct access to charging stations or vehicle interfaces to bypass authentication and billing systems. By leveraging these paths, free-ride attacks compromise economic integrity, leading to revenue losses and undermining user trust in EV charging infrastructure.

4.3 Malware and Phishing/Social Engineering Attacks

Both Malware and Phishing/Social Engineering attacks aim to compromise system integrity by introducing malicious elements, either through software vulnerabilities or human manipulation. While malware targets technical components via malicious code, phishing relies on deceiving users to initiate these breaches. Together, they represent complementary vectors in the cyber threat landscape; one exploiting system weaknesses, and the other exploiting human behavior. This grouping highlights how both technical and social entry points can be used to infiltrate AV, CV, and EV ecosystems.

Malware Injection. Malware injection cyberattacks on EVs and charging infrastructure can exploit vulnerabilities in firmware, software, communication channels, and data handling processes [61]. The attackers aim to inject malicious code into the system, gain unauthorized access, disrupt operations, or compromise the safety and privacy of EV owners through infection of charging station software, malicious software updates, charging infrastructure back-doors, charging mobile apps, EV firmware modification, charging data manipulation, charging infrastructure remote control, unauthorized access to EV data, ransomware, and firmware manipulation for battery overcharging. Malware injection can be caused through any medium to attack all components of the EV system. The consequences of these attacks can affect EV users, service providers (EVCSs), manufacturers, and the overall EV sector [163]. Similar malware injection threats exist for AVs and CVs, which often rely on OTA updates for firmware and software patches.

Malware injection attacks utilize both the A8 and A10 attack paths. Through A8, attackers compromise distributed algorithms and remote communication channels, deceiving systems into accepting malicious software as legitimate updates or commands. Meanwhile, A10 represents physical manipulation vectors, where direct access to charging stations or vehicle interfaces is exploited to introduce malware. By leveraging these paths, malware injection attacks threaten the safety, efficiency, and economic viability of EV operations and broader CV ecosystems.

Phishing and Social Engineering. With the increasing integration of mobile applications, online user portals for locating/navigating the charging stations, and remote management systems in EVs and charging infrastructure, phishing attacks can exploit human vulnerabilities rather than abusing technical flaws. Attackers can create fraudulent websites or send deceptive

communications to deceive EV owners, charging station operators, or maintenance personnel into revealing credentials or installing malicious software. Given the interdependence of EVs and power grids, phishing attacks might have detrimental impacts on power grid as well [149]. Such tactics can also apply to AVs and CVs, where driver or operator accounts are used to manage navigation and OTA updates.

Phishing and social engineering attacks operate through the A9 attack path, which targets human–machine interactions by exploiting social engineering techniques. Rather than attacking technical components directly, attackers manipulate user behavior, leading to unauthorized access, data breaches, and system compromises. By leveraging A9, phishing attacks undermine system safety, compromise sensitive data, and cause financial losses for users and service providers alike.

Summary of this section.

— In the I2D stage, transportation systems involving AVs, CVs, and EVs are vulnerable to various cyberattacks, including impersonation, MitM, Sybil, free-ride, malware injection, and phishing/social engineering attacks.
— These attacks exploit vulnerabilities across communication and control systems, human–machine interfaces, and software components, leveraging attack paths such as A8, A9, and A10. Their impact ranges from unauthorized access and manipulation of system operations to degradation of user safety, operational efficiency, economic performance, and data integrity.

5 Attacks During the Decision to Actuation (D2A) Stage

In this section, we examine cyberattacks that target decision execution and control functions during the D2A stage of AVs, CVs, and EVs. These attacks are grouped based on similarities in how adversaries exploit vulnerabilities in vehicle actuation and control mechanisms, making systems susceptible to disruptions that compromise safety, degrade performance, or interfere with normal operations. For each attack, we present a concise description and specify the associated attack paths (A10 to A11), as depicted in Figure 1. We then assess the potential real-world impacts, particularly how these attacks may affect system performance in terms of safety, efficiency, and economic reliability. Table 5 presents the summary of cyberattacks associated with the D2A stage.

5.1 Software and Infrastructure Manipulation Attacks

Software and Infrastructure Manipulation attacks are grouped together as both involve tampering with critical system components either internal to the vehicle or external within the supporting infrastructure. These attacks aim to disrupt control, communication, or decision-making processes by altering software logic or manipulating the broader operational environment. Grouping them emphasizes their shared capacity to undermine vehicle behavior and system-wide reliability, whether through direct software compromise or indirect manipulation of infrastructure that CVs depend on.

ECUs Software Flashing. The attack involves the malicious alteration of the ECU software. Attackers introduce tampered software into the ECUs via the V2I medium, primarily affecting the control and communications components. This can lead to changes in vehicle behavior, reduced efficiency, or malfunctions. An example is an attacker altering an AV's fuel consumption software, leading to operational inefficiencies or complete breakdowns. Khemissa and Urien [88] proposed a centralized architecture for managing the security of ECUs in **connected and autonomous vehicles** (**CAVs**). The study introduced a lightweight session key agreement scheme based on

Table 5. Cyberattacks During D2A Stage

Attack Type	Cyber Attack	Definition	Attack Paths	Affected Metrics	Related Works
Software and Infrastructure Manipulation	ECUs Software Flashing	Alter ECU firmware (via V2I) to impair control/performance.	A8, A10	Efficiency	[88]
	Infrastructure	Attacks urban traffic systems to disrupt transport coordination.	A11	Efficiency	[95]
Denial of Service and Jamming	Denial of Service (DoS)	Floods EVCS or vehicle networks to halt services/updates.	A10	Safety, Efficiency	[55, 112, 155, 167]
	Jamming and Interference	Disrupts wireless V2X or EVCS communication with noise signals.	A10	Safety, Efficiency	[42, 54]
Tampering, Overpower and Side-Channels	Tampering	Includes malicious firmware/data updates or BMS manipulation in EVs.	A10	Safety, Efficiency	[45]
	Overpower	Forces high voltage through EVCS to damage grid/components.	A10	Safety	[17]
	Side-Channels	Exploits electromagnetic leakages to extract keys or protocols.	A10	Safety, Economy	[22, 50, 94]

symmetric cryptography, enabling secure communication between each ECU and the manufacturer's data center. The approach utilizes a random nonce concatenation operator, a basic hash function, and a keyed-hash message authentication code to enhance security within the CAN bus network.

Such attacks exploit the A8 and A10 attack paths. Through A8, attackers manipulate remote software update processes or distributed control algorithms, injecting malicious firmware into ECU systems via compromised communication channels. A10 reflects physical access-based vectors, where attackers directly flash ECUs using onboard interfaces or compromised charging infrastructure. By leveraging these paths, ECU software flashing attacks compromise vehicle safety, degrade operational efficiency, and pose significant risks to both vehicle functionality and user trust.

Infrastructure. This attack targets the infrastructure supporting CVs. By compromising systems that CVs rely on, primarily through V2I and V2N mediums, it affects the communications component. The potential outcome is systemic failures or large-scale disruptions. An example is a large-scale attack on a city's traffic management system, disrupting numerous CVs and causing significant traffic congestion. La Fontaine et al. [95] analyzed the feasibility of manipulating vehicle routes in metropolitan traffic systems through alternative route-based attacks. This study demonstrated that an attacker could effectively disrupt traffic flow, force vehicles onto sub-optimal routes, or target specific vehicles for various harms, by manipulating road segments in real traffic networks.

Infrastructure-based attacks exploit the A11 attack path, which focuses on external factors influencing system operation. Unlike attacks that target in-vehicle components, A11 attacks manipulate environmental and infrastructure elements, indirectly affecting vehicle decision-making and control. By compromising supporting systems, attackers can induce cascading effects across CV networks, degrading operational efficiency, compromising safety, and overwhelming urban mobility systems.

5.2 Denial of Service and Jamming Attacks

Denial of Service (DoS) and Jamming attacks are grouped together as they both aim to overwhelm vehicular systems by disrupting the communication and control channels essential for real-time operation. While DoS attacks flood systems with excessive requests at higher network

layers, jamming attacks interfere with signals at the physical layer. Despite targeting different levels of the system stack, both attack types compromise the execution of control functions during the D2A stage, ultimately degrading service availability, system responsiveness, and operational safety across AV, CV, and EV environments.

Denial of Service (DoS). In the D2A stage, CVs are also threatened by DoS attacks, which overwhelm vehicle systems and impair their operational functionality. The method primarily involves inundating CV systems with excessive and unnecessary requests. Leveraging the V2N medium, it targets the control and communications component. The attack's consequences are severe, as CV systems can be rendered inoperative. An illustrative example is an attacker sending multiple false traffic updates to a CV's navigation system, causing continuous rerouting and rendering navigation unfeasible. Mohd et al. [112] focused on the simulation and analysis of **Distributed DoS** (**DDos**) attacks on CAV networks. This article used the OMNET++ simulation tool to model and evaluate the impact of DDoS attacks on the network's performance and stability. Yeasmin and Haque [167] proposed a novel approach to defend against DDoS attacks targeting OTA update processes in CAVs using the Hyperledger Fabric blockchain. The proposed **Collaborative Distributed DoS Attack Defense** (**CDAD**) architecture allows multiple OTA service providers to collaborate in real-time to protect against potential cyberattacks. They demonstrated that this collaborative approach ensures the integrity of data and minimizes the risk of malicious actors disrupting the OTA update process while offering improved privacy and security. DoS attacks on EVs and EVCSs affect normal functioning and accessibility of the system, network, or service (e.g., vehicle malfunction), as well as user trust, travel behavior, mobility, and financial benefits. DoS attacks can leverage V2V, V2I, V2N, and V2S to hamper battery, controller, wired, and wireless charging components [55, 155]. In the case of EVs, CAN Bus flooding, battery management system overload, OTA flooding, and mobile-app or remote control flooding are possible attacks. On the other hand, charging station overload, authentication and authorization exhaustion, malicious firmware updates, communication protocol exploitation, and remote management interface overload are mechanisms to attack charging infrastructures.

DoS attacks utilize the A10 attack path by targeting the actuation and control execution processes. Unlike attacks that tamper with data interpretation, A10 exploits physical and interface-level vulnerabilities to flood systems with malicious requests, exhausting computational and communication resources. This results in delayed or failed actuator commands, degraded mobility services, and compromised system efficiency and safety.

Jamming and Interference. Jamming attacks can disrupt the wireless communication channels that are used by EVs, charging stations, and back-end networks that support the vehicle/charger infrastructures by broadcasting interference signals with specific frequencies [54]. These disruptions can prevent EVs from receiving critical updates or authorization tokens for charging transactions, and they can undermine the coordination of V2V or V2I communications. In the context of EVs, these attacks can disrupt battery health monitoring and scheduling of charging sessions. For instance, a jamming attack can disrupt the communication between onboard sensors such as those monitoring battery status and the ECU, ultimately preventing EVs from transmitting this critical information to the server [42].

Such attacks exploit the A10 attack path by targeting the actuation layer through physical-layer interference. Jamming attacks impair the ability of system components to execute or respond to control commands, effectively causing system paralysis. By leveraging A10, these attacks degrade both safety and operational efficiency, resulting in service disruptions, coordination failures, and increased risks in vehicle and infrastructure operations.

5.3 Tampering, Overpower, and Side-Channels Attacks

Tampering, Overpower, and Side-Channels attacks are grouped together as they all exploit physical-layer vulnerabilities during the actuation and execution stage (D2A) of vehicle operations. These attacks bypass traditional network defenses by targeting hardware interfaces, control systems, and unintended signal emissions to alter system behavior or extract sensitive information. Despite differing in technique whether through direct manipulation, electrical overload, or covert signal analysis, they share a common reliance on physical proximity or access to disrupt safety, performance, and reliability in AVs, CVs, and EVs.

Tampering. In EVs and EV charging infrastructure, tampering cyberattacks pose serious risks to safety, functionality, and user trust [45]. Some of the attack mechanisms specific to this domain include charging station manipulation, malicious software updates, data falsification, vehicle software manipulation, BMS exploitation, supply chain attacks, and physical tampering. Such an attack can be carried out through the V2S medium to target all of the EV components. In a broader context, tampering may also manifest in other vehicular systems by manipulating firmware or data exchanges that are crucial to the operational integrity of AVs and CVs.

Tampering attacks leverage the A10 attack path by directly interfering with actuation and control execution processes. Unlike network-layer attacks, A10 exploits physical access or close-range vulnerabilities to alter system behavior at the hardware or low-level software level. This results in compromised operational integrity, degraded system efficiency, and heightened safety risks for users and infrastructure alike.

Overpower. An overpower cyberattack on EVs and charging infrastructure involves manipulating the electrical systems to introduce excessive electrical current or power beyond safe operating limits. Due to the interdependent nature of power and transportation systems, such attacks can lead to cascading failure of components in both systems [8]. The attack mechanisms in this context can vary depending on the specific targets and vulnerabilities present in EVs and charging infrastructure [17]. Some of the attack mechanisms specific to this type of cyberattack include charging infrastructure manipulation, smart charging system exploitation, remote control vulnerabilities, malicious firmware updates, charging authentication bypass, smart grid vulnerabilities, and overpowering electric grids or vulnerable IoT devices. Such attacks can be carried out through V2I and V2N to hamper wireless charging.

Overpower attacks utilize the A10 attack path by interfering with actuation and control mechanisms, specifically at the physical and electrical interface level. By introducing unsafe power levels, attackers compromise hardware integrity, disrupt safe operation, and risk damage to vehicle systems, charging stations, and even the broader power grid. Exploiting A10, overpower attacks pose severe safety risks and can result in extensive operational disruptions across transportation and energy infrastructures.

Side-Channels. Side-Channels cyberattacks on EVs and EV charging infrastructure can exploit the unintended information leakage from the physical implementation of the systems involved [94]. This attack can target signals or patterns unintentionally emitted by various components of these systems, e.g., electromagnetic radiation from a charging controller or power consumption patterns. By analyzing these side-channels, attackers can infer sensitive information, manipulate charging protocols, or extract behavior of EV users in terms of charging. For instance, in case the adversarial actor can gain access to the FPGA controller of the onboard charger, it can initiate a side-channels attack on the residing logic blocks [22]. This attack can leverage V2I and V2N mediums to damage any of the EV system components.

Side-Channels attacks leverage the A10 attack path by exploiting physical-layer vulnerabilities during actuation and control execution. Unlike attacks focusing on software or network-level

manipulation, side-channels attacks use indirect observation and analysis of physical signals to extract or influence critical data. By targeting the interface between hardware operations and system control, these attacks compromise user safety, breach data confidentiality, and threaten the economic viability of EV services.

Summary of this section.

— In the D2A stage, cyberattacks such as ECU software flashing, infrastructure manipulation, DoS, jamming and interference, tampering, overpower, and side-channels attacks pose critical threats to emerging road transportation applications involving AVs, CVs, and EVs.

— These attacks exploit various communication mediums (e.g., V2I, V2N, V2S) and target system components such as controllers, batteries, charging units, and communication interfaces. By leveraging attack paths like A8, A10, and A11, they disrupt system operations, degrade safety and operational efficiency, compromise economic reliability, and reduce user trust and service availability.

6 Existing Research Methods

Ensuring the safety, security, and reliability of AVs, CVs, and EVs requires rigorous testing and evaluation of their resilience against potential cyber threats. This section aims to explore the prevalent methodologies used to test and evaluate cyberattacks on these systems by addressing the following research question: **What methodologies, frameworks, and metrics are currently used to analyze and evaluate cybersecurity in AVs, CVs, and EVs, and what are the significant findings, benefits, and constraints of each methodology?** This research question aims to deliver a comprehensive analysis of the methodologies employed to test and evaluate cyberattacks targeting AVs, CVs, and EVs. Generally, we found three primary methods from the literature: qualitative analyses, field experiments, and simulations.

6.1 Qualitative Analyses

Qualitative analyses use commonly accepted **cybersecurity frameworks** (**CSF**) to derive the threat models and their potential impacts. For example, Bakhtina and Matulevicius [13] identified the protected assets and a threat model in AVs based on a domain model for **information systems security risks management** (**ISSRM**). In [172], the authors unveiled the possibilities of large-scale data harvesting using "nearby cars" features of ride-hailing companies based on the corresponding public documents. Kaja [84] used the Microsoft STRIDE framework to analyze threats in CVs. Similarly, Shirvani et al. [145] established a comprehensive framework for evaluating security risks by first formulating EV security profiles and then mapping the corresponding vulnerabilities to the STRIDE threat taxonomy.

Qualitative threat modeling methods offer many advantages over simulations and field experiments, particularly in terms of cost-effectiveness, safety, and breadth of analysis. Unlike simulations and experiments that require costly infrastructure, qualitative approaches can be carried out early in the design phase with minimal resources, allowing proactive security planning before implementation [146]. These methods also pose no operational risk, avoiding the potential damage of equipment or vehicle accidents. Finally, qualitative models provide broad threat coverage, including logical, policy-based, and sociotechnical risks that are difficult to reproduce in simulations. Our proposed three-stage framework can further enhance qualitative analyses by systematically

categorizing cyberattacks along defined attack paths (A1–A11) across S2I, I2D, and D2A stages, and by assessing their impacts on safety, efficiency, and economy.

While qualitative methods can serve to formulate a high-level assessment of the risks, they have several limitations. They inherently lack empirical validation against real-world adversarial behaviors which reduces their reliability when predicting attack effectiveness or system resilience [29]. Further, due to the static nature of most threat models, they often fail to capture the dynamics of adaptive adversaries or the performance of dynamic defensive strategies, which are crucial in vehicular cybersecurity scenarios. Also, qualitative approaches may oversimplify the complex interplay between cyber-physical components, leading to incomplete risk assessments in safety-critical systems [23]. Finally, while qualitative frameworks highlight possible vulnerabilities, they typically may not prioritize risks based on severity. This limits their utility for risk management and mitigation planning in operational settings.

In addition to technical progress, future research must deploy qualitative and human-centric methods to address the multi-dimensional cybersecurity challenges in AV/CV/EV deployments. Solely quantitative or tool-driven approaches may not properly capture organizational, human, and process-oriented vulnerabilities. Hence, approaches such as ethnographic studies and expert elicitation can provide further insights. Threat modeling workshops are also considered as effective qualitative tool. They bring together cross-disciplinary teams (e.g., manufacturers, suppliers, ethical hackers, transportation agencies, and other stakeholders) in multiple sessions. These workshops systematically explore and discuss attack scenarios on a vehicle or infrastructure systems [62]. Via frameworks such as STRIDE or TARA, participants can jointly identify potential threats, impacts, and mitigation strategies. These collaborative modeling approaches not only produce more comprehensive security requirements but also build a shared understanding and security mindset across traditionally independent domains. By designing an iterative process between technical solutions and qualitative feedback loops, the community can develop more practical and accepted cybersecurity strategies for the future transportation systems.

6.2 Field Experiments

Field experimentation remains the most direct and reliable method for evaluating the effectiveness of cyberattacks and corresponding defense mechanisms on real-world systems. By testing under realistic operating conditions, field experiments provide invaluable insights into system vulnerabilities and the efficacy of mitigation strategies. However, this approach often faces significant challenges, including high costs, logistical complexity, and scalability limitations, which can hinder comprehensive testing at the intended attack magnitude. Despite these challenges, field experimentation is pivotal for identifying vulnerabilities and potential threats associated with AVs, CVs, and EVs across all three stages of the cyber-attack lifecycle (S2I, I2D, and D2A). Effective vehicle cybersecurity testing demands dedicated facilities equipped with the technical and technological capabilities necessary to examine multi-layered vulnerabilities across AVs, CVs, and EVs, as well as their associated infrastructures, such as charging stations, roadside communication or processing units, and traffic control systems. Numerous CV testbeds are available both in the U.S. and internationally, offering controlled, instrumented, or real-world environments for the design, deployment, and evaluation of V2X technologies. Prominent CV testbeds in the U.S. include the **Clemson University Connected Vehicle Testbed** (**CU-CVT**) [26], the Mcity test facility [106], the California CV testbed, the Anthem CV testbed [19], and the NYC CV Pilot Deployment site [122]. These facilities enable V2X connectivity using various wireless technologies, including 3G, 4G, **Long-Term Evolution** (**LTE**), 5G, Wi-Fi, **Dedicated Short-Range Communication** (**DSRC**), and **Cellular V2X** (**C-V2X**). Most of these testbeds also support broker-enabled and message-based communication, facilitating seamless interaction between connected entities within the CV ecosystem.

While these facilities primarily target connectivity testing, many can support cybersecurity evaluations related to V2X communication. Beyond these, there are specialized testbeds focused exclusively on automotive cybersecurity, employing simulation-based, hardware-based, or hybrid approaches. Noteworthy examples include the **Open Car Testbed and Network Experiments** (**OCTANE**) [39], the cyber assurance testbed for heavy vehicle electronic controls [33], the hardware-in-loop-based automotive embedded systems cybersecurity evaluation testbed [125], and the **portable automotive security testbed with adaptability** (**PASTA**) [156]. These testbeds support the testing of diverse attacks, such as message injection, message sniffing, DoS, replay, code injection, and GPS spoofing, across various in-vehicle attack surfaces, including OBD-II, USB, and Ethernet ports, infotainment gateways, and navigation systems [102].

Although the U.S. and other nations host various pilot deployment sites and testing facilities to assess the efficacy, safety, and security of AVs, CVs, and EVs, formal implementation guidelines for state and local agencies remain largely undeveloped. While these pilot studies have generated valuable insights into the safety and operational benefits of AVs, CVs, and EVs, they have yet to provide detailed design guidance for critical infrastructure components, spanning from transportation to digital infrastructure, on how to conduct cybersecurity testing and develop cybersecurity-related certification strategies.

To address this gap, it is imperative to prioritize the development of comprehensive testbeds capable of evaluating both vehicular and infrastructure safety, security, and performance. These testbeds may be physical, virtual, or hybrid, and emulate real-world scenarios with high fidelity. Public-private partnerships will be instrumental in the construction and operation of these advanced testing environments. Furthermore, a coordinated international effort is necessary to establish robust certification strategies for AVs, CVs, and EVs, ensuring that safety, security, performance, and interoperability standards are consistently met. To achieve this, trade organizations, government agencies, and key stakeholders must collaborate to design comprehensive certification frameworks. Establishing clear testing and certification protocols is vital not only for safeguarding public safety but also for accelerating the widespread adoption of these transformative technologies.

6.3 Simulations

Simulations occupy a valuable middle ground between the vagueness of qualitative analyses and the costly (and narrow) field experiments. Simulations enable controlled and repeatable testing of security scenarios with more realism than qualitative methods, but at lower cost and operational risk than real-world deployments. In particular, simulations help evaluate not just isolated vulnerabilities, but also the system-level impact of attacks on safety, functionality, and resilience. Simulations thus provide a safe, cost-effective, and scalable way to validate defenses in diverse driving conditions, attacker behaviors, and operational constraints.

Although simulation is the most widely employed approach in this field, it is essential to recognize that not all simulators are equally credited. To provide clarity and distinction among various simulation-based studies, we focus on the data generation capabilities of simulations, highlighting the corresponding problem addressed by these simulations. The first branch of studies uses simulation as a sensor data generator, addressing the S2I stage of the cyberattack lifecycle [77, 82, 87]. The second branch utilizes simulation as a communication data generator, which involves the I2D stage [15, 28, 35, 76, 127]. The third branch employs simulation as a traffic and infrastructure data generator, requiring the consideration of all three stages of the cyberattack lifecycle i.e., S2I, I2D, and D2A [25, 56, 74, 85, 85, 104, 165].

Despite advances in cybersecurity research for emerging road transportation technologies, there remains a notable deficiency in high-fidelity simulation environments. Current simulations often

lack realism and fail to accurately replicate complex cyber threats and their interactions within transportation systems. This limitation hampers thorough testing and validation of cybersecurity measures under realistic conditions, potentially leaving vulnerabilities undiscovered until they are exploited in real-world scenarios. High-fidelity simulations are essential not only for proactive threat assessment and mitigation but also for enhancing the robustness of cybersecurity strategies at all stages of transportation systems, from streaming to actuation. Addressing this gap is critical to strengthening the resilience of next-generation road transportation technologies against evolving cyber threats.

Advancing cybersecurity for transportation also heavily depends on next-generation simulations and digital twins to evaluate and train systems in complex adversarial scenarios. It may not be feasible to expose real road traffic to every cyberattack due to the potential detrimental consequences. Hence, researchers are leveraging high-fidelity virtual environments to model both the cyber and physical aspects of vehicular systems while minimizing the negative impact of exploring cyberattacks' impact on the real transportation systems [78]. One promising approach is the use of digital twins which are virtual replicas of vehicles (or entire transportation ecosystems) that mirror the state of their real-world counterparts in real time. For example, a digital twin of an autonomous car can be subjected to a spoofed GPS signal or a CAN bus DoS in simulation, and engineers can observe the downstream effects on the car's localization or braking behavior without causing risk to actual passengers [62]. In addition to static simulations, co-simulation platforms are emerging as a promising solution. They combine multiple domain simulators (e.g., traffic flow, vehicular network, power grid, and communication networks) to capture the interdependent nature of smart road systems [133]. For instance, the integration of traffic simulators (e.g., SUMO) with network simulators (e.g., OMNeT++ or ns-3) and vehicle controllers in frameworks such as VEINS. These co-simulation platforms allow researchers to model and inject a cyberattack and observe its impact on network performance and traffic safety outcomes across the I2D and D2A stages. This has specifically been very impactful in the field of AV testing.

Summary of this section.

— Qualitative analyses, field experiments, and simulations are the major approaches for studying cybersecurity issues in AV, CV, and EV systems.
— Among these approaches, simulation is the most widely used, which, based on the simulation purposes, can be further categorized into three branches: simulate sensor data generator, simulate communication data generator, and simulate traffic/infrastructure data generator.

7 Discussions

In this study, we leverage the proposed S2I–I2D–D2A taxonomy to systematically examine how adversarial actors exploit vulnerabilities across the streaming, decision-making, and actuation stages of vehicular operations in AVs, CVs, and EVs. Our analysis elucidates both the technical mechanisms underlying these attacks and their tangible real-world consequences, including operational disruptions, compromised safety, and diminished user trust, factors that collectively threaten the widespread adoption of next-generation transportation systems. Additionally, we identify critical gaps in current evaluation practices and highlight the need to expand the cybersecurity discourse to include the cyber-social domain, where user behavior and societal perception play pivotal roles. The subsequent subsections detail stage-specific vulnerabilities,

outline targeted defense strategies, and discuss broader implications for resilient system design, policy formulation, and regulatory action.

7.1 Key Findings in S2I Stage

In this stage, adversaries primarily exploit vulnerabilities in sensors, communication channels, and data-exchange protocols. One significant observation is the prevalence of manipulation and interception tactics directed at sensor data, which can lead to erroneous perceptions of the driving environment. For example, remote sensor attacks that alter LiDAR readings or intercept signals from TPMSs highlight how even minor manipulation of sensor outputs can result in dangerously incorrect vehicle responses. In addition, communication networks themselves face direct threats. Routing-based intrusions, such as black hole or gray hole attacks in vehicular networks, can reroute or drop packets, leading to data leaks or complete DoS scenarios. Similarly, location trailing and GPS spoofing attacks raise privacy concerns by enabling real-time tracking of vehicles and misleading navigation systems.

Another area of concern involves short-range communication exploits and replay vulnerabilities that leverage proximity-based channels, such as the CAN. By capturing and retransmitting valid signals, attackers can bypass elementary security measures and gain unauthorized control of vehicle functions. Data falsification amplifies this vulnerability by allowing adversaries to feed corrupted or misleading sensor inputs into decision-making algorithms. When these algorithms rely on false data, vehicles may be prompted to engage in abrupt maneuvers, hazardous braking, or other actions that endanger both passengers and surrounding traffic. Taken as a whole, S2I Stage underscores the importance of robust encryption, secure authentication, and multi-layer validation mechanisms to preserve the integrity and confidentiality of data that informs subsequent vehicular decisions.

7.2 Key Findings in I2D Stage

During the I2D stage, attackers shift focus to manipulations that can subvert or distort the logic and control algorithms of CVs. Impersonation attacks illustrate how unauthorized entities can pose as legitimate servers or modules to inject malicious firmware or data. Such impersonation compromises a vehicle's ability to discern authentic instructions, potentially leading to erroneous control decisions. Software compromise also remains a critical risk. Malware injection and phishing campaigns target firmware updates or user credentials, especially in contexts where OTA updates are common.

In EVs and their charging networks, Sybil and free-ride attacks highlight a further dimension of risk, wherein large numbers of fictitious nodes generate confusion or unaccounted resource usage. By injecting numerous counterfeit identities, attackers can exploit charging processes, degrade network trust, or abuse payment systems. This undermines not only the financial viability of charging services but also erodes user confidence in the overall ecosystem. Accordingly, I2D Stage demonstrates that securing the decision-making pipeline demands robust authentication protocols and continuous integrity checks, ensuring that vehicles and chargers can identify and isolate malicious data, nodes, and actors.

7.3 Key Findings in D2A Stage

In the D2A stage, vehicles execute the very decisions that guide their driving and charging behaviors, making this phase particularly susceptible to interventions with immediate real-world consequences. Software manipulations, such as ECU software flashing, can directly alter ECUs or other vital modules, creating conditions for sudden loss of control, operational inefficiencies, or even deliberate shutdowns. Attacks on supporting infrastructure, including traffic-management

systems or large-scale charging networks, highlight how malicious actors can cause cascading disruptions and widespread congestion. By tampering with traffic signals, for instance, adversaries may throw entire transportation corridors into disarray.

DoS and jamming attacks also pose notable threats at this stage, as they can overwhelm communication channels and hamper essential updates or safety-critical messages. When vehicles cannot receive or process timely information such as last-minute warnings or navigation changes, safe action actuation becomes difficult or impossible. Additionally, overpower and side-channels cyberattacks underscore how attackers can push electrical or electromagnetic parameters beyond safe limits to damage batteries, degrade charging infrastructure, or glean sensitive information from inadvertent leaks. Overall, D2A stage reveals the importance of robust safety assurances, intrusion detection mechanisms, and multi-layer redundancy. By ensuring that compromised instructions or power surges trigger failsafe responses, transportation networks can maintain resilience against malicious interventions at the final step of vehicle operation.

7.4 Impacts and Uncertainties in Social Space Due to Cyberattacks

Despite extensive research on **cyber-physical-social systems** (**CPSS**), the social dimension remains largely underexplored, resulting in a lack of standardized models that fully integrate cyber, physical, and social spaces [168, 169]. Recent studies, however, are increasingly emphasizing the need to address this gap by incorporating social components, recognizing that humans act not only as users but also as dynamic sensors and system entities. A key objective of CPSS is to utilize social behavior data and relationship analytics such as sentiment analysis, to provide proactive, high-quality, and personalized services [160]. In the context of AVs, CVs, and EVs, existing reviews have predominantly focused on cyberattacks impacting the cyber-physical domain, often overlooking their broader cyber-social implications [22, 38, 81, 153]. However, cyberattacks across the three-stage workflow, S2I, I2D, and D2A, can have cascading effects that permeate both domains.

At the S2I stage, attacks such as remote sensor attacks, GPS spoofing, and replaying compromise the integrity of raw data collected from the environment. These attacks can alter perception, misguide situational awareness, and provide unauthorized access. While their immediate impact manifests in the cyber-physical space through navigation errors or traffic disruptions, their cyber-social implications include erosion of user privacy, financial fraud through misinformation, and reduced confidence in the system's reliability. In the I2D stage, attacks like impersonation and phishing/social engineering target the interpretation and decision-making layers. Impersonation attacks deceive systems into accepting malicious entities, while phishing exploits human vulnerabilities to extract sensitive information. Though these attacks primarily affect the cyber-physical space by enabling unauthorized control or data breaches, their broader cyber-social impact involves loss of personal data privacy, financial exploitation, and a decline in public trust, which can hinder technology adoption. At the D2A stage, disruptive attacks such as DoS, jamming and interference, and ECU software flashing directly affect system execution and control actuation. These attacks can lead to service outages, degraded vehicle performance, and even safety-critical failures. Beyond these immediate cyber-physical consequences, they influence the cyber-social domain by undermining user trust, altering travel behavior, and inflicting reputational and financial damages on service providers.

In summary, while cyberattacks at each stage (S2I, I2D, D2A) have distinct technical impacts within the cyber-physical domain, their long-term repercussions are profoundly felt in the cyber-social space. These include diminished user trust, compromised data privacy, financial losses, and reduced willingness to adopt AV, CV, and EV technologies. Addressing these multifaceted impacts necessitates a CPSS framework that integrates cybersecurity strategies across all three stages, with an explicit focus on both system resilience and societal acceptance.

7.5 Current AI/ML Techniques and Emerging Research Directions

Current AI/ML Techniques. Modern road transportation systems increasingly leverage AI/ML tools at each stage of the S2I–I2D–D2A pipeline to improve cybersecurity. At the S2I stage, ML-based anomaly detection methods evaluate sensor inputs and in-vehicle networks for potential malicious deviations in real time [131]. For instance, recent studies deployed hybrid rule-based and ML classifiers on CAN bus data to flag intrusions with over 90% accuracy while taking into account the latency constraints [131]. In practice, automotive IDS that use such algorithms have demonstrated detection of over 90% of known attack scenarios in CVs, such as Garrett's IDS [44]. At the I2D stage, **reinforcement learning** (**RL**) can be applied to vehicular decision systems to improve their resilience and adaptability. RL-based controllers can learn to make safe driving decisions (e.g., lane changes or obstacle avoidance) even under anomalous conditions by training against simulated adversaries [63]. Authors in [63] developed a robust RL approach to handle worst-case perturbations in sensor inputs and dynamics by learning an online adversary model. At the D2A stage and across multiple stages of the S2I-I2D-D2A pipeline, secure **federated learning** (**FL**) techniques enable vehicles and infrastructure to collaboratively train models (for anomaly detection, traffic prediction, critical infrastructure resilience) without a need to share the local data of each client (e.g., each vehicle) with a global server [68, 69]. Notably, a recent survey highlighted that FL has been used for tasks ranging from cooperative perception to motion planning in CAVs, with security measures such as secure aggregation and trusted execution environments to tackle challenges such as model poisoning [24]. While this article provides a rigorous review of AI/ML techniques in cybersecurity, our focus is on exploring the potential impacts of cyberattacks on road transportation systems, rather than providing a detailed technical analysis of these methods. For a comprehensive review of AI/ML-based cybersecurity methods, we refer readers to [126].

Emerging Research Directions. Focusing on emerging research areas, there are some trends of developing advanced AI paradigms to address the limitations of today's data-driven algorithms and to ensure security across the vehicle lifecycle. One promising avenue is neuro-symbolic AI, which integrates the learning capabilities of neural networks with the structured reasoning of symbolic systems. Neuro-symbolic systems aim to combine the pattern-recognition power of deep learning with human-like logical inference [70]. This potentially can allow an AV to infer the continued presence of a temporarily hidden pedestrian by applying common-sense rules (object permanence) and ultimately improve safety against corner-case attacks that may confuse classic neural models [70]. Preliminary studies have shown the potential of integrating symbolic logic into deep RL for enabling safe policy learning that avoids unsafe actions in real driving scenarios that can ultimately achieve faster convergence than conventional RL [142]. Another emerging trend is **explainable AI** (**XAI**) for transportation systems. There is a growing trend that black-box ML models in safety-critical AV/CV/EV applications need to be more interpretable [93]. Emerging XAI techniques (e.g., interpretable neural nets) aim to help engineers and policy-makers understand an AI's decisions for example, explaining why an autonomous car's perception module misclassified an object or why an intrusion detector raised an alert [93]. Together, these emerging concepts (e.g., neurosymbolic reasoning to embed safety rules and XAI to provide visibility) offer the potential to enhance future S2I–I2D–D2A stages. They can enable fleets of AVs/CVs/EVs to reason about complex scenarios (e.g., novel attack patterns), explain and justify their defensive actions.

7.6 Stage-Wise Design Recommendations in the S2I–I2D–D2A Framework

Building on our proposed survey framework and taxonomy of attack paths, Table 6 outlines key design recommendations aimed at improving the cybersecurity of emerging transportation

Table 6. Design Recommendations in the S2I–I2D–D2A Framework Across Cyber–Physical Attack Path

Attack-path stage	Key design recommendations
S2I: Stream to Information	• Employ sensor redundancy and cross-validation to counter raw-data manipulation (e.g., GPS spoofing, LiDAR tampering). • Secure data-transmission channels with encryption and authentication to defeat interception, replay, and injection. • Strengthen edge preprocessing via anomaly detection and semantic plausibility checks before data reach central processing.
I2D: Information to Decision	• Implement context-aware decision verification leveraging map constraints and expected driver behavior. • Harden decision pipelines through end-to-end integrity checks and trusted-execution environments. • Integrate model explainability to facilitate rapid anomaly identification and post-incident diagnostics.
D2A: Decision to Actuation	• Design resilient actuation pathways with safety fallbacks allowing safe override or graceful degradation. • Continuously monitor actuation commands and actuator states to detect malicious manipulation. • Deploy adaptive failsafe mechanisms that respond proportionally to jamming or DoS disruptions.
Cross-stage	• Adopt standardized evaluation metrics (safety, efficiency, compromise rate) for reproducible assessment of attacks and defenses. • Develop modular, high-fidelity simulation environments capturing cyber-physical-social feedback loops. • Allocate defensive resources using threat-informed criticality analyses of components and links.

systems. These recommendations are organized around the three-stage feedback loop—S2I, I2D, and D2A—and are informed by common attack patterns and performance evaluation gaps identified in the literature. Besides, these recommendations provide a roadmap for system designers, researchers, and policymakers to build more secure, resilient, and trustworthy transportation technologies, grounded in a structured understanding of system stages, attack mechanisms, and real-world impacts.

7.7 Implications for Government and Public Sector

Many cybersecurity frameworks have been developed by the government and public sector for securing AVs, CVs, and EVs. ISO 27001 and ISO/IEC 27002 provide a foundational **Information Security Management System (ISMS)** for protecting organizational assets by addressing confidentiality, integrity, and availability through organizational, personnel, physical, and technical controls [71, 72]. These standards are globally recognized, offering best practices and supplemental controls based on an organization's specific risk assessments. ISO/SAE 21434 targets automotive cybersecurity, guiding risk management throughout a vehicle's lifecycle [73]. It prescribes methods for identifying, prioritizing, and mitigating cyber threats using standardized terminology (e.g., asset, threat scenario, vulnerability) and a defense-in-depth strategy, along with governance processes to maintain continuous risk assessment across the supply chain. On the other hand, NIST's CSF 2.0 is a flexible risk-management model centered on three components—Core, Profiles, and Tiers [124]. Its Core organizes activities into six functions (Govern, Identify, Protect, Detect, Respond, Recover), enabling structured gap analyses and improvement plans. Because it

is technology-agnostic, automotive applications map seamlessly onto it (e.g., using "Identify" for in-vehicle sensors and "Detect" for intrusion detection).

MITRE's ATT&CK is a comprehensive taxonomy for threat modeling, originally developed for industrial systems but applicable to vehicular contexts [30]. Organized in a matrix of adversarial tactics (e.g., reconnaissance, privilege escalation) and techniques, it also recommends mitigation measures, such as sandboxing and network intrusion prevention. In contrast, **Oak Ridge National Laboratory**'s (**ORNL**) framework is a software tool that simulates vehicle-level cyberattacks on interconnected subsystems, particularly those using the CAN bus [119, 123]. By modeling the entire vehicle rather than individual components, it reveals systemic vulnerabilities and facilitates comprehensive cybersecurity strategies for manufacturers and stakeholders.

Cybersecurity and Infrastructure Security Agency (**CISA**)'s Guidance provides a framework for assessing cyber-physical threats to autonomous ground vehicles, featuring an **Autonomous Vehicle Cyber-Attack Taxonomy** (**AV|CAT**) for categorizing risks by attack vectors, targets, and consequences [32]. This guidance emphasizes a multi-layer mitigation approach, promoting communication and collaboration to address both enterprise- and asset-level vulnerabilities. On the contrary, FHWA's CARMA Program is an open-source platform for **cooperative driving automation** (**CDA**) research, offering tools and simulation environments for testing V2V and V2I communications [40]. Focused on commercial vehicle applications, CARMA serves as a cybersecurity testbed by examining potential attacks (e.g., DoS, message spoofing) and validating intruder detection systems, ultimately fostering collaboration among government, academia, and industry [16, 100, 101, 147].

We have grouped the frameworks into two; one consists of the frameworks developed specifically for vehicle security and the other consists of general CSFs that can be adapted for vehicle security. Several of the frameworks address the cybersecurity risks and the various risk mitigation techniques. Specifically, they discuss various steps involved, including (i) identifying threats, (ii) detecting the attacks, (iii) developing techniques to address the attacks, and (iv) reducing and minimizing the impact of the attacks. For example, the CISA framework identifies the cybersecurity risks in vehicles, detects the attacks (e.g., attacks to supply chain and sensors), and subsequently utilizes risk mitigation techniques such as design, develop, and implement cybersecurity standards.

The ORNL framework also identifies the threats, detects the attacks (e.g., attacks to the CAN), and then utilizes techniques to prevent and recover from the attacks. The tool developed by ORNL helps the defender of the system to conduct penetration testing and gather data to detect vulnerabilities and subsequently develop solutions to address these vulnerabilities. ISO/SAE 21434 framework also identifies the risks and subsequently mitigates the risk for road vehicles. Finally, the Industry framework developed by IIC focuses mainly on safety regulations and only addresses security with respect to protecting information.

Some of the generic frameworks we have discussed can also be adapted for vehicles. For example, the NIST framework (CSF) consists of multiple functions including Identify, Protect, Detect, Response, and Recover. These functions identify the threats, develop controls, detect attacks, apply techniques to respond to the attacks, and finally, recover from the attacks to minimize the damage caused by the attacks. The MITRE Framework also focuses on the tactics used by the attackers, the techniques for each of the tactics, and solutions to mitigate the risks.

Some of the frameworks address different aspects. For example, the framework developed by the **Autonomous Vehicle Industry Association** (**AVIA**) focuses on the regulations for safety, and while security aspects such as protecting confidential information are addressed, much of the focus is on safety considerations. The **SMM** (**Security Maturity Model**) discusses the steps that need to be taken from the current security posture to the desired security posture. The ISO 27001/2

frameworks focus on the various types of controls that relate to organization, people, physical, and technology.

With respect to the three-stage attack lifecycle that we have discussed in this article, the frameworks can provide guidance on the steps to take. From identifying the threats to mitigating the attacks. For example, consider the frameworks such as those from CISA, NIST, and MITRE. At each stage of the lifecycle, the frameworks provide guidance to detect and model the threats, detect the attacks, and mitigate the risks. For example, at each stage, the NIST framework will identify the threats, detect the attacks resulting from the threats, respond to the attacks, and recover from the attacks. The ORNL tool, which is based on its framework, can be applied to gather data at all stages of the lifecycle and identify vulnerabilities and develop solutions. To apply the ISO frameworks, we need to map the controls to the three stages of the life cycle. For example, for the stage S2I, we have to determine which controls are relevant and apply them to respond and recover from the attacks. With respect to the SMM framework, we have to identify the current security posture, the desired security posture, and subsequently develop solutions to migrate from the current to the desired state. Therefore, if the current posture has identified, for example, 10 attacks and the desired posture aims to reduce this number to three, steps must be taken to eliminate seven of those attacks.

7.8 Implications for Industry and Private Sector

The AVIA's Federal Policy Framework outlines legislative and regulatory guidelines to advance safety and innovation in AVs [11]. Key points include extending coverage to all vehicle types, preserving federal authority over vehicle design and performance, protecting confidential information to reinforce cybersecurity, and expanding access to AVs for those without a driver's license. The framework also highlights the importance of technical standards development, research, and data sharing, while discouraging policy delays unrelated to AV deployment.

Qualcomm has prioritized security in its V2X protocols with embedded cryptographic mechanisms. Qualcomm explains how Snapdragon™ C-V2X chipsets embed a hardware root-of-trust that stores IEEE 1609.2 certificate chains and performs on-chip ECDSA verification, thereby eliminating the classic data-falsification (A5) and impersonation (A8) vectors at the S2I gateway [132]. A complementary white paper, C-V2X Drives Intelligent Transportation, highlights Qualcomm's secure key-provisioning service and rapid OTA patch mechanism, which narrows the attack window for remote-sensor tampering or replay-based spoofing (A2/A3) identified in our taxonomy [34]. These measures demonstrate how an S2I-centric cryptographic envelope can be extended through trusted-execution domains to protect the I2D decision logic that Intel's Mobileye EyeQ5 SoC hardens with an integrated hardware-security module [110] and that emerging automotive domain-controller architectures secure by deploying in-chip trusted execution environments and formally verified isolation mechanisms [49, 109].

To enable systematic, large-scale evaluation of vehicular security countermeasures, Qualcomm researchers released **V2X Application Spoofing Platform** (**VASP**), an open-source VEINS module that implements 68 basic safety message manipulation attacks spanning from A1 to A8 [9]. The attack set spans the entire control pipeline, ranging from perception-layer manipulations, such as speed-inflation and phantom-vehicle message injections that compromise the S2I stage, to control-layer incursions, forged emergency electronic brake light and intersection movement assist alerts that subvert the D2A chain. Further, European leading companies, such as Bosch, contribute by developing in-vehicle intrusion detection and advocating for ISO/SAE 21434 and UN Regulation 155 compliance [111, 140, 141]. The mentioned efforts along with other initiatives, align with the S2I–I2D–D2A model, for instance, NIST and Qualcomm address S2I by securing data; Auto-ISAC and Intel's tools strengthen I2D via robust decision logic protections; and

standards such as UN Regulation 155 mandate ongoing actuation monitoring, which ultimately reinforces D2A.

To enhance cybersecurity in emerging road transportation applications (AVs, CVs, and EVs), governments and industry stakeholders must prioritize mitigation efforts based on both the severity of attack impact and evidence of real-world exploitation. Severity refers to the extent to which a cyberattack compromises system safety, operational efficiency, or economic vitality, while real-world evidence is derived from documented incidents, such as those reported by the **Center for Strategic and International Studies** (**CSIS**) [21], which highlight practical vulnerabilities observed in operational contexts. Cyberattacks including DoS, phishing and social engineering, malware injection, data falsification, Tampering, infrastructure manipulation, impersonation, GPS spoofing, eavesdropping, and free-ride have been substantiated by real-world incidents and therefore warrant attention. These attack types often exploit high-risk attack paths such as A10 (e.g., jamming, overpower, tampering), A8 (e.g., impersonation, malware injection), and A2/A3 (e.g., GPS spoofing, replaying), targeting critical system components including communication networks, charging systems, and control modules. Given their proven ability to disrupt safety-critical operations, compromise data, and undermine user trust, these vectors should be prioritized for regulatory intervention, cybersecurity investment, and the development of resilient defense mechanisms.

7.9 Emerging Quantum-Safe and AI-Enabled Mitigation Strategies

Attack types such as remote sensor attacks, routing manipulation, location trailing, close proximity and CAN exploits, side-channels, MitM, repudiation, relaying, and Sybil, while not yet widely recorded in real-world transportation-specific cyber incident databases, represent emerging threats with demonstrated risks in simulation-based, theoretical, or cross-domain studies. These attacks often exploit early-stage system functions (e.g., sensing, authentication, localization, firmware control) via pathways such as A1, A9, A10, and A11, and may lead to significant consequences if left unaddressed. Proactive mitigation through quantum-safe cryptographic protocols, AI-enabled anomaly detection, and secure software/firmware architectures will be essential for minimizing their future impact.

Emerging technologies such as quantum computing and AI offer transformative capabilities for securing next-generation transportation systems. Quantum AI techniques are under investigation for their potential to enhance intrusion and anomaly detection in vehicular environments [75, 135]. Moreover, quantum optimization shows promise in enabling faster threat response and system reconfiguration during cyber incidents [18]. As quantum computing evolves, it threatens to compromise widely used public-key cryptographic algorithms like **Rivest–Shamir–Adleman** (**RSA**) and **Elliptic Curve Cryptography** (**ECC**) through Shor's algorithm [103]. This renders traditional cryptographic infrastructures obsolete and necessitates the adoption of quantum-safe cryptography, including lattice-based, hash-based, and code-based schemes [99]. Protocols such as quantum-safe TLS and digital signatures are actively being explored to secure OTA updates, firmware integrity, and credential provisioning in AV, CV, and EV ecosystems [86, 105]. To future-proof systems, government and industry must not only accelerate investment in quantum-resistant technologies but also define interoperability standards and certification frameworks for post-quantum security in transportation.

8 Conclusions

This survey provides a comprehensive synthesis of cybersecurity vulnerabilities, real-world impacts, and evaluation practices in the context of next-generation transportation systems—specifically AVs, CVs, and EVs. Motivated by the increasing convergence of these

technologies and the shared cyber-physical architecture they rely on, we address critical gaps in the literature through an integrated review framework. Guided by three research questions, we analyzed 227 peer-reviewed studies published between 2021 and 2025 from the Scopus database and introduced two novel taxonomies: a three-stage cyber-attack lifecycle (S2I, I2D, and D2A), and 11 identified attack paths that reflect real-world threat vectors across AV, CV, and EV systems. Our findings reveal that a majority of studies focus on vulnerabilities in the streaming and data processing stages, where input data integrity—such as falsified sensor readings or communication messages—is the most frequently exploited vector. While simulation platforms remain the dominant approach to evaluate attack strategies and countermeasures, we identified a lack of standardization in metrics and evaluation frameworks. Furthermore, despite growing academic interest, few studies quantify the physical and operational impacts of successful attacks, such as loss of vehicle control, traffic disruption, or grid instability. This highlights a disconnect between technical vulnerabilities and their consequences in real-world deployments. By bridging this gap, our survey complements industry standards like ISO/SAE 21434 and NHTSA guidelines, offering a detailed academic perspective that supports practical risk assessment and mitigation strategies. We conclude with a set of actionable design recommendations—emphasizing sensor redundancy, decision validation, and resilient actuation—that collectively support a unified and robust cybersecurity posture for AVs, CVs, and EVs.

Future research should aim to develop cross-domain evaluation frameworks that reflect the integrated nature of these technologies and prioritize empirical studies that map technical exploits to real-world safety outcomes. As next-generation road transportation systems evolve, it is imperative that cybersecurity strategies co-evolve to address emerging vulnerabilities and ensure the integrity, reliability, and resilience of these complex, interconnected infrastructures. Besides, other forms of emerging transportation technologies, such as drones, hyper-loops, or e-bikes, can be considered. However, since these technologies serve different functions within the road transportation system compared to AVs, CVs, and EVs, their direct integration into our scope remains limited. Future work could explore potential interactions or complementary roles these technologies may have alongside AV, CV, and EV systems to improve transportation efficiency and sustainability. More specifically, multi-modal transportation systems are anticipated to introduce new attack surfaces due to the integration of various communication protocols, the addition of one or more dimensions when calculating **origin-destination** (**OD**) pairs, and the challenges of interoperability. For instance, designing cyber-defense strategies that simultaneously protect drones and ground vehicles may inadvertently lead to the emergence of more sophisticated cyberattacks targeting multi-modal transportation systems. Further, this requires extensive research efforts to develop cyber-defense mechanisms that are effective across multiple modalities of transportation while capturing the complexities of co-defending more than one modality.

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